Supplementary Material for: A particle model of hexane removal and water uptake in soybean meal desolventizing

Tomáš Svoboda[1,3,*], Marcela Sluková[1]
Tomáš Moucha[2], Phuong Hoang[2]

[1]University of Chemistry and Technology Prague, Department of Carbohydrates and Cereals, Technická 5, 166 28 Prague 6, Czech Republic

[2]University of Chemistry and Technology Prague, Department of Chemical Engineering, Technická 5, 166 28 Prague 6, Czech Republic

[3]Siemens, s.r.o., Digital Industries, Siemensova 1, 155 00 Prague, Czech Republic

[*]Corresponding author: `tomas3.svoboda@vscht.cz`

Purpose and scope. This Supplementary Material explains the parameter sources, balance formulation, comparison inputs and numerical checks behind the particle calculations. The numerical evidence combines operator verification, conservation checks and selected mesh and time-step sensitivities; complete-trajectory refinement remains the next step in assessing convergence.

The two pure-hexane comparisons examine different transport descriptions: the 2008 falling-rate curves use a *separate Fickian shell*, while the 2019 soybean run uses the water-bearing radial particle. Its volume-mean temperature is compared with the local holder thermocouple as a distinct observable. The DT experiment follows the particle through a prescribed steam-rich bath and uses published outlet ranges as industrial context. Experimental assessment of coupled water and hexane removal requires paired component histories; application to an operating DT also requires tray-scale gas and energy balances and independent process observations. The source release matched to these results is described in Sec. S15; a preservation archive with a persistent identifier is being prepared.

Reading conventions. All solvent and water loadings are kilograms per kilogram of dry, water- and solvent-free composite meal. A *declared* input is a chosen, bounded model assumption; a *measured* input comes from a material or process observation. An *identified* coefficient is evaluated against the same series used to obtain it, whereas an independent prediction uses separate observations. Alternative readings of an unspecified mass basis or size-distribution shape are reported separately. The thermocouple measures a local thin-layer temperature, whose position within the particles is unknown. Numbers prefixed S refer to this document; numbers without the prefix refer to the main manuscript.

The equation tables in Sec. S14 collect the balances, supporting closures and source references. They distinguish the solved particle equations from coefficient constructions, diagnostic limits and the separate historical Fickian comparison.

S1 Process, balances and independent property inputs

The regional inventories, fluxes and surface assumptions are defined in main Sec. 2. This section derives the moving-front energy jump and binding calorics, then records the precise property inputs and film convention. Component storage includes every local phase; evaporation and retention redistribute it without creating a component source.

The exact balances over a moving control volume are Eqs. (13) and (14) of the main text. Here u is the total internal energy density, j_k a molar component flux and $J_E = q + \sum_k \bar{h}_k j_k$ the total energy flux, on the same datum as storage. Terms proportional to the boundary speeds are fluxes of swept material; dropping them creates or destroys material, even when a fixed-mesh solver converges. The jump conditions, Eqs. (15) and (16), are the thin-volume limits of these balances. The wet-side *donor* is the material actually swept, not an equilibrium interface trace substituted into only one balance.

The denominator of the front velocity, Eq. (17), is this jump written so that the latent heat appears once. Eq. (5) removes the wet-side relative hexane flux, so the dry-side hexane flux is $\dot{s}(C^{\Gamma}_{\mathrm{h,d}} - C^{\mathrm{b}}_{\mathrm{h,w}})$. Substituting it into the energy jump leaves

$$\bar{h}_{\mathrm{h,d}}(C^{\Gamma}_{\mathrm{h,d}} - C^{\mathrm{b}}_{\mathrm{h,w}}) - (u^{\Gamma}_{\mathrm{d}} - u^{\mathrm{b}}_{\mathrm{w}}) \tag{S1}$$

in the denominator. On one caloric datum, $h_v - u_l = \Delta h_{\mathrm{vap}} + (h_l - u_l)$. For the liquid, $h_l - u_l = Pv_l$ is negligible compared with the latent heat. For an ideal vapour, $h_v - u_v = RT/M_{\mathrm{h}}$, so $u_v - u_l = \Delta h_{\mathrm{vap}} - RT/M_{\mathrm{h}}$. The volumetric denominator therefore contains the latent heat of the liquid hexane swept by the front, a smaller ideal-gas correction on the pore-gas trace, and the binding-potential difference already stored in u. The smooth-region balances have no separate vaporization source, so this heat is not counted twice.

S1.1 A shared donor, retained phases and caloric closure

The historical core loading and algebraic total-flux closure are defined in main Sec. 2.2; their heat-fed front limit is explained in main Sec. 2.3. The separate Fickian shell of Eq. (S2) instead assumes a slow transport path that the binary total-flux closure does not supply.

At a moving interface, the water and energy donor terms both use the state of the consumed wet-core material. A necessary discrete datum test is $\partial R_E/\partial c_{\mathrm{w}} = R_{\mathrm{w}}$, with R_E and R_{w} the interfacial energy and water residuals and c_{w} an arbitrary shift of the caloric zero of water. When a stored bulk water term is mixed with a swept equilibrium trace, the test exposes a hidden pre-drainage assumption. The resolved moving-front comparisons in the main text use the corrected donor; both prescribed-bath experiments use the same radial closure and conservative transitions between regimes. The binding energy of retained hexane comes from the same Guggenheim–Anderson–de Boer (GAB) activity law that sets the stored solvent. With sorption potential $\psi_{\mathrm{h}} = (RT/M_{\mathrm{h}}) \int_0^W \ln a_{\mathrm{h}}(\xi, T)\, \mathrm{d}\xi$, the caloric potential is $B_{\mathrm{h}} = T\partial_T \psi_{\mathrm{h}} - \psi_{\mathrm{h}}$. No separately fitted heat-of-sorption source enters the energy equation. The modified Luikov water isotherm, Eq. (30), gives the retained-water activity within its measured activity and temperature domain. Condensing water on a surface exposed to vapour and retaining water in the meal are different operations: condensation changes the material inventory, and retention defines a constitutive state function.

S1.2 Parameters, source hierarchy and film definition

This subsection records the source conventions behind the adopted parameter set, Table 2 of the main manuscript, which separates published property fits, derived quantities and declared transport inputs. The 2008 thesis and 2019 article are each used for their own runs, and a printed table takes precedence over a digitized plot. The 2019 samples' mean sphericity is not claimed as measured for the 2008 samples. Each published correlation has its own definition of gas velocity, voidage and transfer area; borrowing a coefficient without them would obscure the comparison.

The cross second virial coefficient of the binary gas uses the corresponding-states correlation of Plyasunov and Shock (2003), with central interaction $k_{\mathrm{wh}} = 0.494$ and source interval 0.477–0.511.

Both completed main trajectories use the central value; listing the interval does not mean that the full coupled histories were recalculated at its endpoints. The upper water capacity is the modified Luikov value at source-domain activity 0.799, 0.235670904 kg/kg dry; the main table shows it rounded to 0.236. The reference lower moisture anchor is 0.0230 kg/kg dry, and the adopted positive-water continuation has no lower moisture floor. It retains $A_1 = 0.880$, $A_2 = 12.184$ and the analytic log activity of main Eq. 30; exactly absent water is a separate limit. Tests confirm bitwise identity within the original measured band. The reference retained-water mobility is $8.4741855104 \times 10^{-7}$ mol/(m s), constructed in Sec. S12 and held fixed in the pore study. The reference meal heat capacity is 2317 J/(kg K), shown as 2.32×10^3 in the main table; these display choices do not alter the calculation inputs. The derived dry-meal density used in the calculation is 1159.65 kg/m^3; it is shown as 1.16×10^3 kg/m^3 in the main table.

The bed-film correlation is

$$\mathrm{Nu}_\varepsilon = 0.6941 \mathrm{Re}_\varepsilon^{0.579} \mathrm{Pr}^{1/3}.$$

Faner (2008) prints 0.6941; the value 0.6949 transcribed by Coletto, Bandoni, and Blanco (2022) is not used.

The bed groups are $\mathrm{Re}_\varepsilon = \mathrm{Re}/(1-\varepsilon)$ and $\mathrm{Nu}_\varepsilon = \mathrm{Nu}\varepsilon/(1-\varepsilon)$, with holder voidage ε and approach-velocity Reynolds number based on equivalent diameter. Whitaker's separate packed-bed correlation (Whitaker, 1972) is evaluated at each 2008 rig state; the duty scores under both mass conventions are in Sec. S3. The 2019 boundary instead uses the constant-rate exposure construction of Sec. S7.2. The oil anchoring and fluid calorics follow main Sec. 2.8; the older additive oil-arm floor remains a separate sensitivity bracket (Sec. S10).

S2 Conservative numerical method and observed accuracy

This section supports the main text's numerical method and verification. The finite-volume scheme uses one swept wet-volume rate, Eq. (33), for the wet cells, interface and dry cells. The front cuts a fixed set of material shells; grid-motion fluxes at the faces cancel in pairs when inventories are summed. Backward-Euler stepping solves the regional storage changes and the interface's component and energy jumps as one nonlinear system. The event at which the external liquid disappears is found from mass and energy conservation together, with event time and temperature as unknowns. An exactly dry core loses its degrees of freedom instead of being carried as an infinitesimal artificial core. Each event is checked with its corresponding interface equations. State acceptance requires positive phases, a front that does not grow and non-negative holdups; an infeasible trial leaves the previous state unchanged.

Figure S1 separates time advancement from nonlinear iteration. Physical closures are first evaluated at the initial guess to prepare the residual and its scaling. At each subsequent trial state, state-dependent properties, phase equilibrium and transport fluxes are re-evaluated before forming the coupled water, hexane and energy residuals. Surface-water and retention-capacity states follow the selected equations; liquid-water imbibition requires a film-connected dry shell. Conservative event transfers are included in the balances checked before accepting the state. Numerical recovery follows the declared controls of the relevant stage.

The manufactured-solution tests measure the convergence order of *one operator* at a time. The complete particle histories require their own refinement study. Every accepted step closes the water, n-hexane and energy balances, per step and cumulatively. The last three rows of Table S1 examine a constructed numerical stress schedule: over 0.225 s, the external temperature and composition are stepped and then restored, while the front moves only 2.3 % of the radius. This schedule probes the response to abrupt boundary changes and differs from the physical histories of the laboratory and

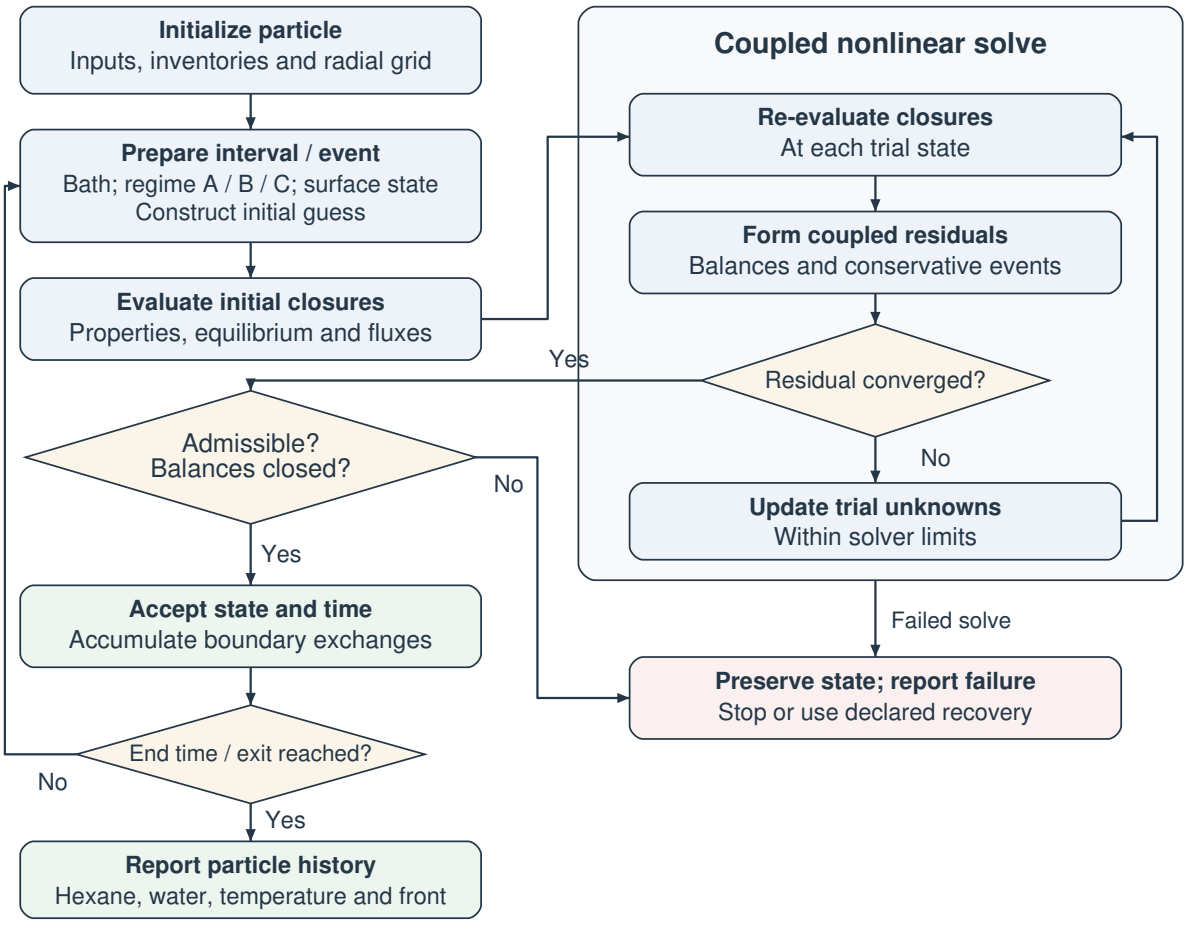


Figure S1: High-level algorithm for the prescribed-bath particle solver.

DT experiments. The manufactured-solution and balance rows above it use separate operator and conservation tests.

S2.1 State acceptance and numerical refinement

For the numerical reference tests, the scaled nonlinear residual and physically normalized conservation bounds are 10^{-10} per step and cumulatively. The activation subsystem keeps its separately declared 2.5×10^{-10}. At an exact grid face, the departure tangent admits an inward root under a declared 1.0×10^{-10} bound, never an outward-moving front. When a feasible state or event cannot be found under these equations and balance criteria, the calculation retains the input state and reports the failure. The refinement ladder uses radial counts $12, 24, 48, 96$ and steps $0.075, 0.0375, 0.01875$ seconds on the same $0.225\,\mathrm{s}$ stress schedule. Adjacent differences must decrease and the finest pair must agree within one percent ($0.1\,\mathrm{K}$ in temperature). Front position and inventories pass this test on that schedule (Table S1).

The 2008 dry-shell curves have a different numerical test: the *surface-read* variable-coefficient march (Sec. S4) is second order in shell mesh (observed 2.00–2.12) and first order in step (1.10–1.16); the reported residuals are within one percentage point of their continuum extrapolations. The current water-bearing experiments continue on a stationary radial particle after a conservative projection removes the core. Their history-specific checks (Secs. S5 and S12) quantify selected mesh

Table S1: Numerical checks and their scope.

Quantity	Observation	Interpretation
Fixed-domain manufactured tests	Space 1.95–2.00, time 0.97–1.00	Separate wet/dry operator verification
Moving-front time operator	Time 0.98–0.99	Formal first-order temporal discretization
Balance for short ladder	Bound 10^{-10}, largest 7.8×10^{-13}	Normalized conservation in the numerical test
Geometric conservation identity	Exactly zero	Shared swept-volume quantity
Caloric-datum check	3.42×10^{-10}	Under separately stated environment allowance
Front on 12–48 cells	Nine of ten three-level chains contract, finest pairs within 0.20 %	Six chains need the declared face-residual bound
Pulse component-flux integrals	No contracting chain; final-pair miss up to 14.8 %	Constructed pulse schedule
Final hexane inventory, same ten chains	Finest pair within 0.40 % on every chain	Same constructed pulse schedule
Cumulative boundary hexane outflow	One-percent pair on nine of ten chains; worst pair 4.8 %	Same constructed pulse schedule

and time-step sensitivities. Component and energy conservation are verified at each accepted step; establishing spatial and temporal convergence requires refinement of the complete histories.

S2.2 Controls for the prescribed-bath histories

The histories use a uniform regime-A body with a separate surface temperature, followed by four radial cells for Faner and two for DT. Root residuals, numerical seeds and terminal-time estimates are reported separately from the normalized component and energy acceptance limit 10^{-10}. Accepted-state records retain the incoming state, source hashes, controller settings and conservation checks; refused attempts remain available with the calculation records (Sec. S15).

Faner roots and core extinction. The existing ordinary-root controller permits residuals up to 10^{-8}; the updated positive-core target solves use the regional 10^{-9} criterion where the stricter root stalls. The dedicated vanishing-core mode permits 2×10^{-8} only for incoming wet volume fractions at most 10^{-7}, with a scaled condition-number ceiling up to 10^{18}. Mesh-face departure seeds may use 10^{-9} and advance no time. Component and energy acceptance remains 10^{-10}, with phase and regime-sequence checks unchanged; tighter root convergence is required when needed to satisfy conservation.

Core extinction follows four actual shrinking-core intervals. Duration contraction supplies an initial guess only; projection still transfers inventories and external exchanges over its estimated duration. The remaining-time bound assumes a stated future contraction ratio and is below 20 ms for Faner (2 ms for DT), rather than a full-history error bound. The updated Faner bound is 10.08 ms, with the last four durations contracting by factors 0.68244, 0.68222 and 0.68205. Repeated targets caused by decimal rounding are refused under the unchanged conditioning ceiling, then replaced by the next prescribed target without changing stored geometry.

Faner's independent whole-history exterior-rate accumulation uses a reporting allowance 4×10^{-10}, corresponding to at most 10^{-11} kg water/kg dry at initial water 0.025 kg/kg. This allowance is distinct from the per-step 10^{-10} conservation gate. The observed defects are reported with the histories and are far below the loading discrepancy. Earlier mesh and timestep comparisons remain identified as constant-property evidence (Sec. S5).

Earlier constant-property DT recoveries. At $K_\ell = 0.32$ mol/(m^2 s), the first shell interval reused a rejected iterate only as a starting guess for the same problem. At the first mesh face, a tangent residual 2.19×10^{-10} exceeded the earlier 2×10^{-10} seed tolerance, so a seed-only bound 3×10^{-10} was declared. The departure solve used the neighbouring $K_\ell = 0.10$ solution for its initial guess and residual-row scaling; the actual 0.32 state, forces and balances were solved anew. Its accepted 0.05-s step had scaled residual 9.2×10^{-11}. The solution-path check retained 3 000 000 sampled points and all tolerances while its wall allowance increased from 120 to 600 s per step. Mobility factors 100 and 10 000 used these controls.

Factor 10 000 required a separate exception at its first 0.05-s face departure: the newborn shell water-row residual could reach 10^{-9} (actual 5.6×10^{-10}, equivalent to 4.2×10^{-12} kg water/kg dry), while every other row stayed below 10^{-10}. Auxiliary starting guesses were admitted up to 2×10^{-9} before the full solve. An algebraically identical liquid-water force, checked against 80-digit arithmetic, avoided cancellation; the path check allowed 12 million points and 1800 s. This water-row exception is the only relaxed accepted-step row tolerance in those earlier studies; particle-wide component and energy balances still satisfy 10^{-10} per step and cumulatively.

The $K_\ell = 0.05$ case used a neighbouring solution as its first departure guess and, after film depletion, adaptively reduced the regime-C step from 5 to 1.25 s. At $K_\ell = 0.20$, exact supply exhaustion gave a transferred/requested-water ratio $1 + 2 \times 10^{-16}$ through round-off; reordered arithmetic returned exactly one, without a clamp.

Current property histories. The adopted factor-100 DT histories use the archived native controller with the existing same-problem birth-iterate starting guesses and face-departure duration retries. A failed birth advances no physical time; accepted material from another coefficient case is not imported. Native root criteria, three-million-point path budget, 120-s per-step path allowance and component/energy acceptance at 10^{-10} are unchanged. Each DT property alternative starts from the common original feed and retains its own law and cumulative ledgers. The 2.5-s regime-C check reuses only the matching factor-100 regime-B checkpoint. The unchanged Faner property histories retain their previously documented controls. Refused attempts and executed settings remain with each history; results are in Sec. S13. These controls do not establish whole-history continuum convergence.

S3 Experimental sources and the regime comparison

The 2008 thesis (Faner, 2008) and later article (Faner, Perez, & Crapiste, 2019) supply distinct traces for the regime comparison. Their source conventions and the 2019 setup are collected in Sec. S7.1; the versioned scored points and redistribution scope are described in Sec. S15. These pure-hexane records assess solvent removal, while coupled water uptake requires steam-exposed observations.

The constant-rate period tests the external heat supply: the coefficient is evaluated at each run's reported gas state, holder footprint and sample mass. Reading the printed mass as the wet charge, the ratio of calculated to measured duty is 0.82–1.06, and that of crossing times 0.86–1.13. Reading the same mass as dry meal gives 0.53–0.61 of the duty and 1.45–1.70 of the crossing time. The thesis's printed flux convention supports the wet-charge reading but does not give an independently measured film coefficient. Of its three bed correlations, Whitaker's gives the closest coefficient. The bracketing criterion set before the comparison is not met, because the coefficients inferred at two conditions lie 3.7 and 4.2 % below Whitaker's. The departure is retained in the duty comparison. This benchmark examines the external correlation and the supplied X_c; identifying a

hexane diffusion coefficient requires a separate intraparticle transport measurement. The measured boiling-point plateaux separately check the pure-hexane property equation.

The regime test is applied before any transport model is chosen. For each of the four plotted traces: interpolate the time at which the measured loading crosses $X_c = 0.20\,\mathrm{kg/kg}$; average the thermocouple over the first thirty seconds; and interpolate the first rise one kelvin above that average. A plateau that ends after the crossing, at a temperature consistent with solvent boiling, is consistent with a receding liquid-hexane core. A plateau that ends before it, with the temperature several reading half-widths above boiling, is inconsistent with a boiling core at that loading. The diagnostic is the sign, not the precise heat-limited time. Table S2 keeps each trace's own clock and shows that the sign is unchanged for rise thresholds of one to five kelvin. Distinguishing the intraparticle mechanisms requires spatial temperature measurements and controlled run conditions beyond those supplied by the source.

Table S2: Measured plateau end minus critical-loading crossing time (s).

Source/meal	+1 K	+2 K	+3 K	+5 K
Faner (2008), soybean	−34.4	−32.1	−29.9	−22.6
Faner (2008), sunflower	−18.4	−15.9	−13.5	−8.9
Faner et al. (2019), soybean	+41.2	+44.6	+47.9	+54.5
Faner et al. (2019), sunflower	+45.6	+51.4	+57.2	+63.7

A heat-limited estimate supplies the latent heat per unit dry meal through the measured constant-rate film and a growing spherical dry shell. The measured 2019 intervals to $X_h = 0.10$ are 1.3–2.1 times this isolated-particle estimate, and the 2008 intervals 12.8–25 times it: the 2019 records are heat-fed, the 2008 records are not. The water-bearing particle (Sec. S5) and the Fickian shell (Sec. S4) take up these two groups, respectively.

S4 The 2008 falling-rate curves and independent uptake

The scalar Fick description used in the main text's separate comparison with these older, slower drying records is

$$\frac{\partial}{\partial t}[\varepsilon_g c_{h,g} + \rho_{dm,p} W_{eq}] = \frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2 D_{eff}\,\frac{\partial c_{h,g}}{\partial r}\right], \tag{S2}$$

where ε_g is the gas-filled porosity, $c_{h,g}$ the hexane mass concentration in the pore gas, $\mathrm{kg/m^3}$, W_{eq} the local equilibrium retained hexane, kg/kg dry meal, and D_{eff} an effective transport coefficient (Cardarelli, Crapiste, & Mattea, 2002). The equilibrium capacity belongs inside the storage derivative, not inside the divergence as a face mobility. This equation serves only this separate comparison; neither its diffusivity nor its scalar flux replaces the binary particle law in the two main experiments.

The 2008 thesis meal is already hotter than boiling at the reported critical loading. Its falling-rate trajectories are therefore tested against the *separate* Fickian shell of Eq. (S2), not against an unobserved front. The soybean-meal apparent diffusivities of Cardarelli (1998) refer to total stored solvent and need a storage conversion before they enter a pore-gas gradient model; the retained-hexane isotherm of Cardarelli and Crapiste (1996) supplies it. The law used here is regressed on the seventeen soybean coefficients, so applying it to the 2008 sunflower trace is a cross-meal check, not a same-meal prediction. At 50 °C twelve uptake series identify coefficients 8.4–20.1 times the tabulated apparent values, within the model's storage retardation at that temperature. Since those values came from the same uptake curves, the match is an identification, not a new prediction.

At hotter process conditions the retardation approaches unity, so the multiplier inferred from the low-temperature curves cannot be transferred unchanged.

The source's thirty-four uptake coefficients across 50–95 °C define the separate effective law

$$\ln D_{\mathrm{eff}} = \ln A - \frac{E_{\mathrm{a}}}{R_{\mathrm{g}} T} + \beta \ln(X_{\mathrm{h}}/X_{\mathrm{ref}}), \tag{S3}$$

with $A = 3.513 \times 10^{-9}\,\mathrm{m}^2/\mathrm{s}$, $E_{\mathrm{a}} = 0.31 \pm 16.1\,\mathrm{kJ/mol}$, $\beta = 0.959 \pm 0.331$ (95-percent half-widths), and $X_{\mathrm{ref}} = 0.01\,\mathrm{kg/kg}$. The residual standard deviation is 0.410 in $\ln D_{\mathrm{eff}}$. The law is extrapolated 41 K to the thesis drying conditions and read at the *surface boundary loading* once per step, a declared convention. A weighted-mean diffusivity (Crank, 1956) predicts much faster desorption, and local evaluation gave no mesh-refined result on the tested spheres. Numerical convergence of the surface-read convention does not extend to those other implementations or remove the temperature and geometry uncertainty. Table S3 reports root-mean-square (RMS) errors as percentages of the measured loading drop and counts points within the digitization band. Both particle sizes use the same coefficient law without refitting; geometry alternatives and the coefficient's prediction band are reported separately (Table S5).

The declared 2008 volume-to-surface radii are 0.7215 mm for sunflower and 0.666 mm for soybean. Their shape-transfer assumption, its selection after radius screening and the separate geometry sensitivity are specified in Sec. S7.3.

Table S3: Errors and digitization-band counts for the 2008 falling-rate comparisons.

Representation	Sunflower	Soybean
Volume-equivalent thesis sphere	37.9%, 1 of 14	31.0%, 1 of 15
Volume-to-surface thesis sphere	17.2%, 9 of 14	15.6%, 8 of 15
Reported discretization versus extrapolated	17.2% versus 16.7%	15.6% versus 15.2%
Across law's 95% prediction factor	5.4–52.6%	7.7–42.2%

In the surface-read model the curves dry *more slowly* than in either 2008 measurement. The residuals on the declared spheres and the range of faster and slower traces within the prediction band show the remaining discrepancy. Interpreting the radius and diffusivity sensitivities physically requires independent particle-shape and transport measurements. The comparisons support examining a sorption-storage-corrected Fickian path in the earlier samples; resolving their microstructure, process-temperature transport coefficient and differences from the later Faner source requires further measurements. A two-size experiment on one feed, with loading and particle temperature both measured, would help distinguish film-dominated from shell-dominated scaling.

S5 The Faner experiment with a water-bearing particle

This section reports the Faner history and its supporting sensitivities. The complete experimental inputs and observable definitions are collected in Sec. S7.1; the exposure construction is in Sec. S7.2. Constitutive assumptions are specified with the main equations, and numerical controls in Sec. S2.2. The nominal four-cell calculation retains the experimental clock and the constant-rate boundary calibration. Its event times and outlet values are reported in main Sec. 4.1 and Table 3. Water and core volume are shown in Fig. S2. The maximum Ackermann argument is 0.639, within the unit reference band and the declared $|\beta| \leq 3$ study band. The loading discrepancy remains visible; no falling-rate or temperature point is fitted.

Earlier mesh and timestep sensitivity. The following comparisons retain the earlier constant pore diffusivity and oil capacity. They do not establish convergence of the updated temperature-property history.

For the earlier constant-property, smaller-radius case, halving the four-cell step of the stationary dry particle from 2.5 to 1.25 s changed final hexane by 0.24%, water by 0.19% and mean temperature by 0.072 K. These earlier time-step checks concern the 0.808-mm sphere. The earlier constant-property source-mean sphere was compared at two and four cells; because the positive-core step sequences differ, the whole difference cannot be assigned to the mesh alone. For this radius, the two-cell and four-cell core-depletion times differ by 2.0 s; at 240 s, hexane differs by 0.024%, water by 1.3% and temperature by 0.031 K. These comparisons quantify numerical sensitivity at the stated resolutions; complete-history refinement is needed to establish convergence.

The initial water loading was varied over 0.025, 0.05 and 0.10 kg/kg dry meal; regime-A histories are admissible at all three. The new 0.05 attempts at the upper corner and at low water mobility find no admissible leading pore profile by shooting, and the 0.10 leading jump seed at the upper corner does not converge. These bounded searches identify continuation difficulties at the tested inputs. A converged continuation is needed before the corresponding physical response, including any phase change, can be assessed. Robustness to initial moisture and the regime-A water-access assumption remain open. The successful 0.025 charge is not presented as a validated reconstruction of a source experiment whose moisture was not reported.

S5.1 Size distribution and startup interpretation

The source size descriptors and nominal sphere are defined in Sec. S7.1. The size and startup sensitivities below retain the earlier constant $D_b = 6 \times 10^{-10}$ m^2/s and oil capacity 2000 J/(kg K).

An earlier construction used a lognormal number distribution truncated to 0.5–4 mm, with log-width about 0.280 and constraints from the source mean and 1–3-mm fraction. Weighting by d^3 gives mass quantiles of about 1.50, 2.15 and 3.03 mm at 10, 50 and 90%. The main calculation uses the source mean diameter as a volume-equivalent sphere. A new coarse size sensitivity uses three bins of equal dry mass with mass-mean diameters 1.60, 2.15 and 2.89 mm, using the earlier constant $D_b = 6 \times 10^{-10}$ m^2/s. With the surface-to-volume mapping $R = \psi d/2$, the corresponding radii are 0.593, 0.796 and 1.068 mm. Their positive-core marches approach core depletion at about 49, 76 and 118 s. The spread of core-depletion times illustrates how a size distribution broadens removal. Completing the class continuations through core extinction is needed to construct a charge-weighted curve and assess the effect over the full falling-rate interval. The assumed distribution is constrained by source summaries, not by fitting the measured drying trajectory.

For classes of equal density and oil content, number weights must be converted to dry-mass weights before superposition:

$$q_c = \frac{p_c d_c^3}{\sum_b p_b d_b^3}, \qquad \overline{X}_k(t) = \sum_c q_c X_{k,c}(t).$$

The mean temperature of such an ensemble still differs from a local thermocouple reading. Choosing a distribution to obtain a desired curve shift would not identify the experimental size distribution.

The startup sensitivities below keep the earlier 0.808-mm sphere. The unresolved preheat endpoint is described in Sec. S7.1.

A bounded regime-A calculation with the same particle, bath and exposure gives external-hexane depletion at 41.078 s for a start at the boiling temperature, 41.437 s at an initial 68.5°C and 42.143 s at 68.0°C. This modest subcooling thus shifts that event by about 0.36–1.07 s; it does not produce a 10–20-s shift of the temperature curve. These calculations cover regime A; a complete drying

history would also require continuation through regimes B and C. Starts at 65 and 60°C gave no admissible step in the bounded search of the evaporation-only regime-A equations. A colder start therefore needs a separately examined thermal and phase treatment of the surface before it can give a new full curve; it is not represented by shifting the existing trajectory in time.

S5.2 Source-size and vapour-temperature sensitivities

This subsection reports complete two-cell sensitivity calculations with the same transport coefficients, initial loadings and experimental clock. Replacing the earlier 0.808-mm radius with the source-mean 0.885-mm radius moves the 0.20-kg/kg crossing from 41.0 to 46.6 s and lengthens the 0.20–0.10-kg/kg interval from 15.5 to 17.7 s. At 240 s, the temperature falls from 116.0 to 114.3 degrees Celsius. The size change thus improves two observables through a physical change in area per dry mass and transport distance; the remaining discrepancies stay explicit.

A second calculation keeps the source-mean state at its computed core depletion, 90.1 s, and then applies the measured vapour-temperature trace to the same stationary-particle equations. The digitized gas readings are joined linearly; the last value, 117.82 degrees Celsius at 235.058 s, is held to 240 s. The final particle temperature is 113.0 degrees Celsius and hexane is 0.00168 kg/kg dry, compared with 114.3 degrees Celsius and 0.00161 kg/kg in the constant nominal bath. This calculation isolates the response to the measured temperature trace after core depletion; the earlier history retains the nominal bath. Its independent whole-history hexane, water and energy defects remain within 4×10^{-10}.

More water could delay warming through evaporation and heat capacity, but it also changes the phase equilibrium and coupled transport. The water loading of the source charge is not reported; it was dried under nitrogen and stored in air before being re-soaked in hexane. Raising the water only to lower the temperature would need information the experiment does not provide.

If the denominator is truly dry meal, reading total mass loss as solvent loss gives an apparent loading $X_{\mathrm{h}}(t) + X_{\mathrm{w}}(t) - X_{\mathrm{w}}(0)$. Water loss lowers this quantity, so it does not explain the under-predicted late solvent loading. Whether the source used a dry or an as-is denominator is a further uncertainty of interpretation.

A further shell-formation search at 0.050 kg/kg, with central differences and the default finite-volume seed, was stopped when it exceeded its allotted search time; it gives no additional drying trajectory.

Figure S2 shows the calculated retained water in the particle and at its surface, and the liquid-hexane core volume fraction. These supporting quantities were not measured in the source experiment. In short, the calculated particle dries faster than the measured charge in the falling-rate period.

S6 Process interpretation, sensitivity and limits

This section connects the particle results to DT operation. The particle supplies the component and energy source or sink terms needed by a tray model. Coupling those terms to tray-scale gas and meal histories requires accounting for incoming particle loadings and sizes, steam condensation, residence, gas composition, mixing, jacket heat and vapour escape. Transfer of a thin-holder film coefficient to an industrial bed requires the corresponding gas flow, contact area and heat path; the source leaves the local sample heat path and probe position unspecified.

Within its modeled range, at the constant gas state of each separate march, the moving-front formulation is heat-fed: the binary pore mobility separates water from n-hexane but adds no independent resistance to their total escaping flux. For the *separate Fickian dry-shell comparison*, the

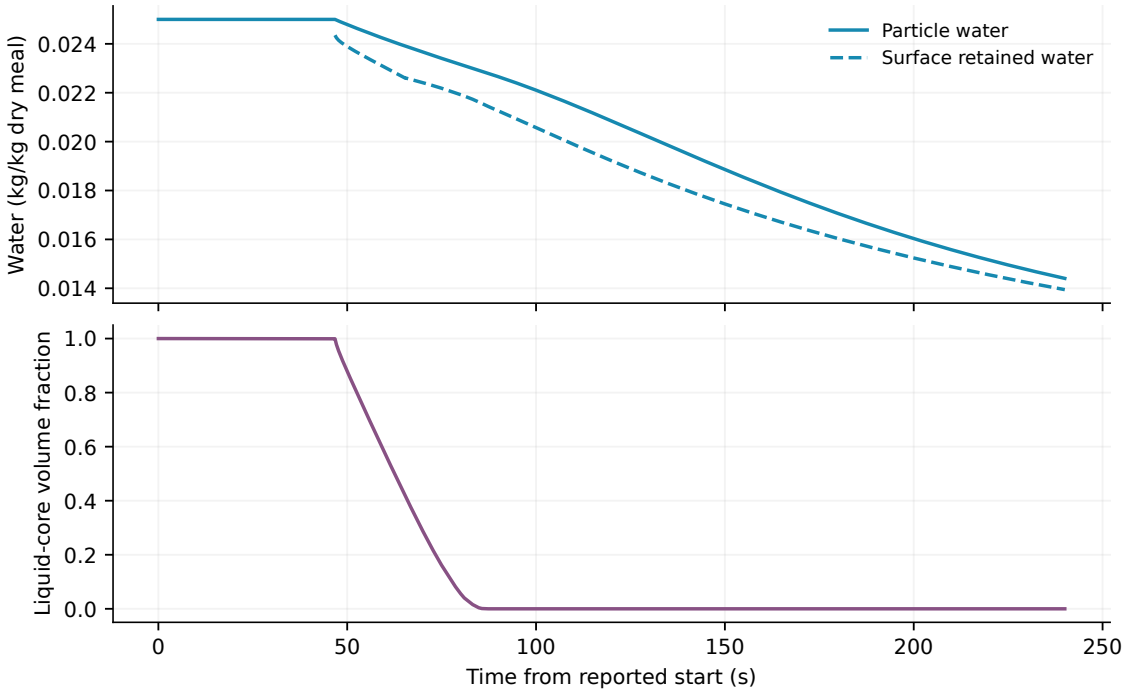


Figure S2: Faner particle: retained water and liquid-hexane core volume fraction.

ratio of conductive heat supply, expressed as an evaporation flux, to diffusive solvent supply is

$$\Lambda = \frac{k_{\mathrm{dry}}(T_{\mathrm{g}} - T_{\mathrm{int}})/\Delta h_{\mathrm{vap}}(T_{\mathrm{int}})}{D_{\mathrm{eff}}[c_{\mathrm{sat}}(T_{\mathrm{g}}) - c_{\mathrm{s}}]}, \tag{S4}$$

where c_{sat} and c_{s} are the n-hexane mass concentrations at saturation and at the particle surface, respectively. On the declared gas-state grid, Λ ranges from 7.80×10^3 to 6.29×10^4 over states with a receding front. A value above unity means that this Fickian comparison is limited by its assumed solvent transport. The ratio characterizes the separate Fickian description at fixed gas states. Assessing the controlling resistance in steam-wet meal requires the coupled heat and component responses along a drying history.

Table S4 summarizes what each comparison establishes and the measurements needed to distinguish the remaining mechanisms (DTDC: desolventizer-toaster-dryer-cooler).

With pressure and source-water-isotherm alternatives propagated, the ideal retained-water vapour-pressure diagnostic $a_{\mathrm{w}}(W_{\mathrm{w}})P^{\mathrm{w}}_{\mathrm{sat}}(T^*) = P$ gives 100.5–107.6 degrees Celsius, a range spanning a published band of industrial discharge temperatures. The calculation assigns the supplied moisture to the retained-water inventory; resolving bound, interstitial and external water separately requires phase-partition measurements. The single point at the moisture of the temperature source itself lies 2.05 kelvin above that band. This cross-source, water-only equilibrium calculation describes retained-water activity. Boiling of a separate free-water droplet depends on that droplet's own liquid state, whereas the coupled particle temperature follows the transient component and energy balances. The binary particle calculation uses water-component fugacity and calculates its temperature independently of T^*. At declared last-tray hexane mole fractions 0.003–0.03 and temperatures 100–110 degrees Celsius, a dry-particle equilibrium calculation gives 4.5–66.0 ppm without and 8.1–112.4 ppm with a residual-oil solution arm (Sec. S10); only the richest and coolest oil-arm corner reaches the published residual-hexane class. These values describe the equilibrium inventory at the supplied conditions; reaching it depends on the transient transport history. The total-meal

Table S4: Scope of the benchmark evidence.

Comparison	Evidence	Further measurements
Heat-fed surface period	Four measured duties under a declared mass basis and an independent packed-bed correlation	Source-specific exposed area and per-run state
Slow Fickian shell	Two 2008 measured trajectories with one surface-read coefficient from uptake	Thesis particle shape, local-coefficient convergence, high-temperature transport
2019 water-bearing particle	A four-cell radial particle with film from the constant rate; falling-rate loading is faster than the measured charge	Measured mass-weighted size histogram, wetness near the probe, its depth and gas flow through the holder
Binary water–hexane storage/transport	Conservation and limiting-case numerical identities for coupled transport	Water-isotherm and co-sorption data; coupled moisture and solvent profiles
Industrial DT/DTDC	One radial particle on a declared bath; solvent, moisture and temperature lie within the published outlet ranges; Coletto et al.'s column integration is not the boundary, because its flashing zone uses eq. A.17 and its toaster particle step has no water balance	Independent tray gas, steam, wall, residence and discharge observations

GAB isotherm already includes reference oil, so assessing the oil-arm overlap requires independent mixture data.

Source-domain limits and constitutive extensions are stated with the main property equations (Sec. 2.8); the front and imbibition assumptions are in main Secs. 2.3 and 2.6. The formulation does not identify shrinkage, a distributed phase change or re-wetting of a receded core. Three groups of transport and surface coefficients have positive intervals without source-backed endpoints. Numerical conservation does not resolve these material uncertainties.

The measurements in Table S4 can be organized as a two-size pure-hexane series with temperature through the crossing, same-meal transport and retention measurements at process temperature, and paired solvent–water histories under a known steam supply. Holder geometry, exposed area and probe wetness must accompany those observations. Equipment assessment additionally needs independent feed, gas, steam, wall-duty, residence and discharge records, followed by the dryer-cooler and whole-tower residence distribution.

S7 Experimental inputs and comparison conventions

This section collects the experimental inputs and boundary construction. Scoring and uncertainty conventions are specified in Sec. S8.

S7.1 Source conventions and Faner setup

Within a source, printed tables take precedence over digitized positions, and a later peer-reviewed condition over an earlier thesis statement. For the 2008 thesis the nominal mass is read as wet charge because that interpretation reproduces the printed flux convention; the dry-mass alternative is evaluated separately (Sec. S3). Version-two digitization uses fourteen sunflower points from the measured crossing. The earlier 9.1% residual on the 2019 sunflower sphere was 14 of 15 points in band on version one; version two gives 11.6%, 13 of 14. This is a changed reading, not a changed model.

For the 2019 soybean run, Faner et al. (2019) supplies pure hexane vapour at 393.15 K and 101325 Pa. The source reports preheating in the apparatus and no visible warming stage, with an initial plateau near 68.7 °C, but gives neither the exact preheat endpoint nor initial moisture. The flakes were desolventized in nitrogen, crumbled, stored at ambient conditions and re-embedded in hexane; their water loading cannot be inferred from fresh extractor meal. The declared body starts uniformly at the calculated boiling temperature, 341.903356 K, with hexane 0.475 and retained water 0.025 kg/kg dry meal, oil fraction 0.0195 and recorded forcing timescale 29.749225 s. Water remains stored but has zero surface exchange during regime A under the access assumption in main Sec. 2.5.

The image-analysis mean equivalent-sphere diameter is 1.77 mm, mean sphericity 0.74, diameter range 0.5–4 mm and fraction between 1 and 3 mm 95.1%. The nominal radius 0.885 mm uses that mean directly, without adjusting it to the drying curve. It preserves equivalent volume and has less area per volume than the flakes; the alternative $R = 0.74 \times 1.77/2 = 0.655$ mm preserves area-to-volume ratio but not internal transport distances. Neither mapping identifies the unreported mass-weighted histogram. Its assumed distribution and earlier size/startup sensitivities are given in Sec. S5.1.

The gravimetric mass loss is read as solvent loss, although water loss was not measured separately. The reported temperature is one local holder thermocouple, while the calculated observable is the particle volume mean. Probe depth, wetness near the probe and run-specific holder gas flow are unreported. These measurement and boundary uncertainties remain distinct from digitization resolution (Sec. S8).

S7.2 External heat coefficient and finite sample geometry

Film exposure multiplier: source data and calculation. The multiplier was inferred from the constant-rate evaporation duty, including the Ackermann correction. For a sphere with a wet surface, $m''\Delta h_{\mathrm{vap}} = h_Q\Phi(\beta)(T_g - T_b)$ and $r_c = 3m''/(\rho_{\mathrm{dm}}R_{\mathrm{p}})$, where r_c is the decline of hexane loading per unit dry mass and $\beta = m''c_{p,\mathrm{v}}/h_Q$. The original construction matched mass-weighted particle rates to a least-squares source slope of 7.26×10^{-3} kg/(kg dry s), using the four soybean solvent-loading readings in Fig. 2 of Faner et al. (2019), above loading 0.2388. The digitized pairs (t_j [s], X_j [kg/kg dry]) are

$$(0.000,\ 0.492364), \quad (9.884,\ 0.420900),$$
$$(20.077,\ 0.337384), \quad (29.961,\ 0.278064).$$

These values keep the digitization precision for reconstruction; it does not represent experimental accuracy. With bars denoting the four-point arithmetic means, the slope is

$$r_c = -\frac{\sum_j (t_j - \bar{t})(X_j - \bar{X})}{\sum_j (t_j - \bar{t})^2} = 7.260603 \times 10^{-3} \text{ kg/(kg dry s)}.$$

The construction used a truncated lognormal number distribution of sizes, consistent with the reported mean diameter and size fractions and converted to mass weights, with reference radius 0.666 mm and initial loading 0.4924. For each size class i, the Ackermann balance gives

$$r_i(f) = \frac{3f h^0_{Q,i}}{\rho_{\mathrm{dm}} R_i c_{p,\mathrm{v}}} \ln\left(1 + \frac{c_{p,\mathrm{v}}(T_g - T_b)}{\Delta h_{\mathrm{vap}}}\right),$$

where $h^0_{Q,i}$ is the unscaled correlation coefficient. The earlier ensemble uses 400 mass-weighted classes. It chooses the single factor f so that the least-squares decline of its mean loading, $\bar{X}(t; f) =$

$\sum_i w_i X_i(t; f)$, at the same four times equals r_c. The smallest classes enter the falling-rate period inside that window; matching only the ensemble's initial rate would give 0.14508. The four-time rule gives $f = 0.1457854364$; the present 0.885-mm particle uses it unchanged, with initial loading 0.475. The multiplier carries a constant-rate boundary calibration from the earlier geometry. Interpreting it as a transport property or exposed-area fraction would require independent flow and contact measurements. The later falling-rate and temperature readings are reserved for comparison. Faner also inferred heat-transfer coefficients from drying data (Faner et al., 2019). The digitized data, distribution inputs, slope calculation and exact carried-over value are kept in a derivation record of the authors, which is not part of the release (Sec. S15) and is available from the corresponding author on request.

The unscaled gas-film heat and mass coefficients use an industrial reference superficial velocity of 0.230 m/s, obtained from a 3.9 kg/s gas flow, a 6-m tower diameter, a gas density of 0.60 kg/m^3 and a bed voidage of 0.40 (Coletto et al., 2022). At the present reference radius, the heat coefficient is 104 W/(m^2 K) and the mass coefficient 0.0672 m/s; scaling gives 15.1 W/(m^2 K) and 0.00980 m/s. Both coefficients are scaled, while the geometric area remains $4\pi R_\mathrm{p}^2$. The inference identifies the heat duty; applying its factor to mass transfer as well is a declared boundary assumption. Faner's charge of about 30 g in a 0.011-m^2 holder forms a layer about three to five particle diameters deep (Faner et al., 2019). Applying industrial reference coefficients to the holder requires a separate assessment of its gas flow and sample contact.

An unscaled calculation, with multiplier one and otherwise the same four-cell particle, initial moisture, oil fraction, transport coefficients and prescribed vapour, depletes external free hexane at 6.8 s and crosses 0.20 kg/kg dry at 6.8 s, rather than 46.8 s and 46.6 s, respectively. Its heat and mass coefficients are 6.86 times larger. The calculation enters regime B and continues to 6.88 s with the same component and energy checks; the bounded continuation then reaches its computing-time limit. This result quantifies the early boundary sensitivity. Continuation beyond the computing-time limit is needed to assess the full falling-rate interval and later temperature.

The independent 2008 duty comparison, its two mass conventions and its narrowly missed bracketing criterion are reported in Sec. S3. Transferring a holder coefficient to a tray requires the tray's gas composition, voidage, interfacial area and condensation balance.

S7.3 The declared spheres and the effective diffusivity law

This subsection details the declared spheres and diffusivity law of Sec. S4. The convention $R_d = \psi d_\mathrm{eq}/2$ gives the sphere the particle's volume-to-surface ratio. It was adopted after the effective-radius sensitivity had been inspected and is an assumption for the 2008 samples, whose shape was not measured; the 2019 sphericity 0.74 comes from later samples. At constant diffusivity the spherical diffusion time scales with $R_\mathrm{p}^2/D_\mathrm{eff}$, so a drying curve alone cannot separate radius from diffusivity. The volume-to-surface equivalence is an early-time construction, while the scored window reaches a Fourier number near 0.3. The law's temperature term is weakly resolved (the 95-percent half-width of the activation energy exceeds its central value), and its prediction factor is a spread within the regression domain, not a statement about the 41 K transfer above it. Reading the coefficient at a weighted mean concentration instead of at the surface would raise it five- to seventy-five-fold and dry far too fast; reading it in each interior cell gave no mesh-refined solution.

Table S5 compares these sphere conventions with the same coefficient law. Errors are RMS fractions of the measured loading drop; parentheses count points within the reading band, out of fourteen sunflower and fifteen soybean observations.

Table S6 tests mesh and time-step sensitivity on each thesis sample's declared sphere. It reports the RMS error of the surface-read Fickian calculation as a fraction of the scored loading drop.

Table S5: Particle geometry sensitivity in the 2008 falling-rate comparisons.

Sphere convention	Sunflower	Soybean
Thesis volume-equivalent diameter	0.379 (1)	0.310 (1)
2019 soybean volume-equivalent diameter, both traces	0.317 (1)	0.303 (1)
ψ times thesis sample diameter, declared	0.172 (9)	0.156 (8)
ψ times 2019 soybean diameter, both traces	0.099 (14)	0.147 (9)
ψ times 2019 sunflower diameter, sunflower only	0.116 (13)	–

Table S6: Space and time refinement of the 2008 Fickian comparisons.

Trace	60 cells, nominal step	120 cells, quarter step	240 cells, quarter step
Sunflower, $R_\mathrm{p} = 0.7215$ mm	0.1721	0.1678	0.1676
Soybean, $R_\mathrm{p} = 0.6660$ mm	0.1562	0.1525	0.1524

Across the law's prediction band, the residual on the declared sunflower sphere runs from 5.4 to 52.6 percent and on soybean from 7.7 to 42.2 percent; the best match lies inside the band, so points inside a broad band do not identify a mechanism. The central marches dry more slowly than both traces. Even so, a storage-consistent path with no coefficient fitted to the falling-rate points comes considerably closer than the same law on the volume-equivalent spheres. The uncertainty is dominated by particle shape and source conditions, not by truncation error in the surface-read form.

S8 Comparison clocks, scores and uncertainty

S8.1 What is compared, and on which time origin

The 2008 Fickian comparison starts at the measured critical-loading crossing. The 2019 calculation retains the reported experimental clock, with no time shift to reduce its falling-rate discrepancy; its initial state and observable definitions are in Sec. S7.1, and its boundary calibration in Sec. S7.2. Crossing times are interpolated linearly between successive states, and figure lines interpolate the saved states. Main Table 3 gives the measured and calculated crossings and falling interval. Numerical controls and refinement evidence are in Secs. S2.2 and S5.

S8.2 Scores and uncertainty conventions

Reading resolution, alternative source interpretations (mass basis, shape, probe position and gas flow), and regression intervals are reported separately; source alternatives are not statistical samples. The band around a digitized point is the reading half-width of this digitization, not an experimental standard deviation; a point *in band* lies within that resolution. The RMS drying score is the root mean square of model minus measured loading over the scored window, divided by the measured loading drop over that window. Changing a class distribution, coefficient or window defines a new calculation, not a new reading of the same score. The regression of converted uptake coefficients states 95-percent half-widths on its exponents and a residual standard deviation in log coefficient; neither covers the thesis samples' geometry or the 41 K transfer. Faner's image-analysis counts give neither a mass-weighted histogram nor the thermocouple's local wetness, and the digitization half-widths represent neither uncertainty.

S9 Property domains and process boundaries

Constitutive domains and the positive-water continuation are specified in main Sec. 2.8. Pure bulk hexane is evaluated separately from the binary pore and surface state, with positive retained water active after shell formation. The source's unreported water loading and the adopted regime-A access assumption are defined in Sec. S7.1.

S10 Gas-composition and outlet-equilibrium calculation

This section gives the equilibrium calculation behind the residual-hexane floor quoted in the main text. The pressure-feasible solvent activity used for the outlet check is $a_{\mathrm{h}}^{*} = \min(1, y_{\mathrm{h}} P / P_{\mathrm{sat}}^{\mathrm{h}}(T))$. The *equilibrium loading* is evaluated at a specified y_h and temperature; the time needed to approach it follows from the transient transport calculation. Table S7 separates GAB-retained solvent from the total dry-shell inventory, which also holds pore gas; its grid samples a literature-based range of last-tray conditions. The last-tray total of 4.5–66.0 ppm without the oil-solution arm refers to $y_h = 0.003$–0.03 and 100–110 °C. With the oil-dissolution sensitivity it becomes 8.1–112.4 ppm; this band must not be formed by adding the arm to a total-meal isotherm as if their overlap had been measured. In the experiments' pure-hexane gas the floor is many times higher. The DT march of Sec. S11 reads this floor at its declared exit gas, $y_h = 0.01$, below the dry-shell tail.

Table S7 uses a pressure of 101.3 kPa. The $y_h = 1$ row represents the experimental pure-hexane gas, not a desolventizer exit. The oil-solution sensitivity (Sec. S6) is not included in the table totals.

Table S7: Equilibrium hexane inventories at atmospheric pressure (ppm dry meal).

	100 °C		110 °C	
y_h	Sorbed	Total	Sorbed	Total
0.003	5.55	6.58	3.51	4.51
0.01	18.52	21.94	11.71	15.04
0.03	55.76	66.02	35.24	45.24
0.10	188.23	222.45	118.90	152.22
1.0	2312.41	2654.60	1416.61	1749.88

Two numerical checks examine the limiting behaviour of the coupled formulation. In the ideal-binary reduction the scalar Fick identity holds to 6.34×10^{-16}, while at finite pressure the low-water coefficient comparison leaves a 3.799 % discrepancy retained in the comparison. The default coupled interface formulation at 409.15 K admits a pore hexane fraction of at most 0.857; this restriction does not define the supplied external bath. The Faner experiment with a water-bearing particle treats the pure bulk gas separately from its binary state in the pores and at the surface; the earlier property-table checks are separate from this march. The experiments that would settle what the comparisons leave open are listed at the end of Sec. S6.

S10.1 Gas-phase flux contributions after core extinction

The transport decomposition uses the outer dry-shell face of the same Faner and DT histories plotted in the main text. For a recorded hexane transfer $\Delta n_{\mathrm{h,out}}$ over a step Δt, the particle-side boundary ledger gives $N_{\mathrm{h}}^{\mathrm{g}} = \Delta n_{\mathrm{h,out}} / (4\pi R_{\mathrm{p}}^{2} \Delta t)$. The recorded gas flux N_{t} excludes retained-matrix water transport. The discrete advective composition is the upstream value $y_{\mathrm{h,up}}$: the outer-cell

value for outward gas motion and the surface value for inward motion. The contributions are

$$N_{\mathrm{h}}^{\mathrm{g}} = y_{\mathrm{h,up}} N_{\mathrm{t}} + J_{\mathrm{h}}, \qquad f_{\mathrm{motion}} = \frac{y_{\mathrm{h,up}} N_{\mathrm{t}}}{N_{\mathrm{h}}^{\mathrm{g}}}, \qquad f_{\mathrm{diffusion}} = \frac{J_{\mathrm{h}}}{N_{\mathrm{h}}^{\mathrm{g}}}.$$

These are signed fractions of net outward hexane flux. Negative values oppose removal; a fraction greater than unity represents an outward contribution partly cancelled by the other contribution. Table S8 gives representative values.

Table S8: Signed contributions to regime-C hexane flux at the outer dry-shell face.

Experiment	Time, s	Total vapour motion, %	Binary diffusion, %
Faner	89.9	100.01	−0.01
Faner	240.0	100.70	−0.70
DT	508.5	97.33	2.67
DT	1766.5	−29.35	129.35

Binary diffusion opposes outward hexane removal in all 62 recorded Faner regime-C advances. In the DT history, total gas motion first turns inward at 613.5 s, turns outward after film depletion (first recorded at 708.3 s), then is inward in all 199 advances from 778.3 s to exit. Hexane continues to diffuse outward during these intervals. The reported clocks are discrete observation times, not localized events marking zero total gas flux. The decomposition identifies the direction and relative magnitude of the two gas-phase contributions. Determining which resistance controls removal requires a separate sensitivity analysis of heat supply, pore diffusivity and retained-water transport.

S11 The prescribed DT bath and particle history

This section records the prescribed DT bath and resulting particle history. Shared constitutive laws and surface assumptions are those of main Sec. 2; tray vapour and energy balances are not solved in this experiment.

Inputs. The prescribed residence is 1766.535501 s and the effective spherical radius is 1.5 mm. This is a selected coarse-particle case; the main text states its origin and distinguishes it from a measured DT mean radius. Feed temperature is 322.15 K, hexane loading 0.387786 kg/kg dry meal, retained water 0.123664122 kg/kg dry meal and oil fraction 0.013740458. The pre-desolventizing (PD) bath is 345.15 K, with hexane mole fraction 0.6644429974, exposure fraction 0.004081 and indirect duty 85 W/kg dry until 490.202242 s. The steam-rich bath then ends at 378.15 K; its gas states and timings are given in the prescribed-bath input file of the release (Sec. S15). These inputs define the prescribed-bath scenario. Comparison with plant dynamics requires measured tray histories or a coupled column calculation.

The common material inputs are in main Table 2; DT imbibition coefficients are constructed in Sec. S12. The coarse spatial representation and numerical controls are specified in Sec. S2.2.

Events and exit. External free hexane is depleted at 492.257071 s, and the first dry-shell interval ends at 492.267071 s. Regime B resolves the front's grid-face arrival and departure before core extinction at 503.540021 s. The external water film stays positive throughout regime B. Regime C advances the stationary radial particle and water film together, resolves film depletion at 703.259973

s and continues to the prescribed exit at 1766.535501 s. Final core volume and external water film are zero. Particle water is 0.211596339 kg/kg dry, hexane is 0.000167695203 kg/kg dry and volume-mean temperature is 378.114756 K.

Wet-basis moisture is $100X_{\mathrm{w}}/(1+X_{\mathrm{w}}+X_{\mathrm{h}})$, giving 17.4618%. Residual hexane is 167.695 mg/kg dry, or 138.389 ppm on a wet basis with the same denominator. This conversion excludes the water stored in the external film. Figure 3 of the main text places the published outlet ranges at 30 min for visual reference; the computed clock and its earlier end point are not shifted. The published ranges place the particle result in industrial context; direct validation requires observations matched to the particle and bath.

Conservation and numerical scope. An independent reconstruction covers every accepted step from dry-shell formation to exit, evaluating water, hexane and energy in the particle and external water film. The maximum normalized defects are 8.31×10^{-12}, 2.37×10^{-12} and 2.34×10^{-12}, respectively. The preceding feed history is regenerated with the integrated oil law; independently reconstructed exterior rates give maximum normalized hexane, water and energy defects 3.76×10^{-13}, 1.11×10^{-13} and 1.89×10^{-13}. The feed is common to the diffusivity sensitivities because regime A contains no binary pore transport. Its inputs and boundary are held fixed. The updated regime-C step check is in Sec. S13; regime-B recoveries and their numerical scope are in Sec. S2.2.

The available checks quantify selected numerical sensitivities; complete-history mesh and time-step refinement remains to be established. The regime-B trajectory retains a positive water film throughout, so assessing the depleted-film branch requires a separate case.

Each segment records the hashes of its source files and of the saved state from which it starts. The updated histories use one isolated native-source snapshot, with an explicit property-law adapter and recorded numerical controllers. The release (Sec. S15) provides sources, inputs and histories for tracing the reported calculation. The balance and numerical checks characterize this discrete particle experiment. Identification of its material coefficients and assessment at equipment scale require independent particle and process measurements.

S12 Liquid-water imbibition: coefficient basis and numerical evidence

This section constructs the water-transport coefficients and reports their sensitivities. The physical pathway, activation conditions, retained inventory and omitted mechanisms are specified in main Sec. 2.6.

The constitutive derivations below remain the basis of the updated DT case. The factor, contact-conductance and earlier timestep tables retain their original constant-property calculations: $D_b = 6 \times 10^{-10}$ m^2/s and oil capacity 2000 J/(kg K). They document earlier sensitivity evidence, not sensitivity or convergence of the updated temperature-dependent case. The updated nominal history and balances are reported in Sec. S11; its property sensitivities are reported separately.

Reference transport coefficients. The current pore law and coefficient-convention limits accompany main Eq. 12. No physical justification has been established for the former scalar-to-binary factor of 1.5.

The reference retained-water mobility was originally derived from the linearized coupled balances at $T = 385.65$ K, $P = 130$ kPa and $y_{\mathrm{h}} = 0.80$, with the reference retention properties. Eliminating the total molar flux gives the local apparent response

$$D_{\mathrm{app}} = D_{\mathrm{app,pore}} + g_{\mathrm{ret}}\mathcal{L}_{\mathrm{w,ret}},$$

where $D_{\mathrm{app,pore}} = 4.54 \times 10^{-12}$ m^2/s and $g_{\mathrm{ret}} = 7.26 \times 10^{-4}$ m^3/mol are obtained from the same storage and chemical-potential derivatives. Setting the combined response to 6.2×10^{-10} m^2/s, with the former constant $D_b = 6.0 \times 10^{-10}$ m^2/s, gives the reference retained-water mobility 8.47×10^{-7} mol/(m s). The target value is the maximum apparent moisture diffusivity reported by Touffet, Déléris, and Bokhove (2026) for pelleted feed at 60 °C; transferring it to this reference state is an engineering assumption. On the isothermal, uncapped Luikov branch, the diffusivity of the retained-water path is

$$D_{\mathrm{ret}} = \frac{M_{\mathrm{w}} \mathcal{L} A_1}{\rho_{\mathrm{dm}} A_2 W^2},$$

which for the reference mobility is 3.8×10^{-10} m^2/s at $W = 0.05$ and 2.4×10^{-11} m^2/s at $W = 0.20$. Compare this with two oilseed-meal datasets: vapour-sorption diffusivities of sunflower meal, 10^{-12} to 1.15×10^{-10} m^2/s at 50–95 °C (Cardarelli, 1998), and the soaking diffusivity of defatted soybean meal, 1.14 to 2.07×10^{-11} m^2/s at 20–60 °C (Mukherjee, Chakraborty, & Dutta, 2019). The reference retained-water mobility has a comparable scale to oilseed-meal measurements, although its value was taken from the pellet study. It is held fixed in the new full-history pore-diffusivity comparisons; the local apparent response is not forced to remain exactly at its original target when the pore law changes.

The liquid-water mobility factor. The selected total enhancement is $F = 100$; the additional path is $\mathcal{L}_\ell = (F-1)\mathcal{L}_{\mathrm{w,ret}} = 8.39 \times 10^{-5}$ mol/(m s). We estimate its scale from the hot-water uptake of soybean meal reported by Hemmingsen et al. (2008), using the coarse mean sieve diameter 1.43 mm as an assumed sphere diameter. The first coarse sample at 80 °C, after 90 s immersion, already has substantial uptake near the later cake-water level. We interpret 90 s as an approximate equilibration timescale and use 95–99% uptake as a declared endpoint convention, not source-measured equilibration fractions. For constant diffusivity, uniform initial loading and an equilibrated surface, the spherical solution (Crank, 1956) requires Fourier numbers $Dt/R^2 = 0.253$–0.416 at these endpoints. With $R = 0.715$ mm and $t = 90$ s this gives $D(80\,^\circ\mathrm{C}) = (1.44\text{–}2.36) \times 10^{-9}$ m^2/s. An assumed inverse-viscosity transfer, using the water-viscosity ratio $\mu(80\,^\circ\mathrm{C})/\mu(100\,^\circ\mathrm{C}) = 1.257$ (International Association for the Properties of Water and Steam, 2011, eq. 7), gives $(1.81\text{–}2.97) \times 10^{-9}$ m^2/s at 100 °C. Dividing by the baseline $D_{\mathrm{ret}} = 2.377 \times 10^{-11}$ m^2/s at the representative loading $W = 0.20$ gives factors 76–125; we adopt the rounded value 100. This selects a constant mobility enhancement from an approximate transport scale; it is not a measured uncertainty interval or an additional temperature-dependent liquid-transport law.

The source cake water includes external and interstitial water, and the approximately 150-s drainage period does not isolate intraparticle uptake. The sphere, 90-s particle-equilibration interpretation and viscosity transfer are therefore engineering assumptions. The finite $K_\ell = 0.32$ mol/(m^2 s) is retained separately; the combined closure is not fitted to reproduce the cake uptake in 90 s. Hong and Lee (2006) distinguish fast network water and slower hydrated water in soft tofu, with molecular self-diffusivities 2.26×10^{-9} and 6.84×10^{-11} m^2/s. This supports distinct molecular mobilities, but self-diffusion is not the present collective transport coefficient and does not set F.

The mobility and contact terms act in series. The following resistance estimates and marches retain their earlier constant-property scope. Table S9 gives the interior share of the series resistance at the DT radius with $d = R_{\mathrm{p}}/5$, the lumped-sphere equivalent, for $K_\ell = 0.32$ mol/(m^2 s). Above a factor of about 300 the surface carries most of this lumped resistance. At 100 the interior approximately doubles it; a larger contact conductance can partly compensate. These resistance estimates do not identify the two coefficients from the cake-water data. Completed DT marches at factors 100, 1000 and 10 000 with the same K_ℓ (Table S10) place the outlet on a plateau between 1000 and 10 000; where between 100 and 1000 the plateau begins was not marched. The tenfold reduction

to 100 lowers the outlet moisture by 0.58 percentage points and raises hexane by 22.8 mg/kg dry; the tenfold increase to 10 000 moves them by only +0.07 points and −2.7 mg/kg dry. The earlier constant-property outlet is on its plateau at 1000; the adopted factor 100 includes appreciable internal resistance.

Table S9: Interior share of the series resistance to liquid entry at the DT radius, $K_\ell = 0.32$ mol/(m^2 s), $d = R_\mathrm{p}/5$; total mobility factor F, added mobility $(F-1)\,\mathcal{L}_{\mathrm{w,ret}}$.

Mobility factor F	Interior share	Uptake slowdown vs. contact alone
30	0.80	4.9
100	0.53	2.1
300	0.27	1.4
1000	0.10	1.1
3000	0.04	1.04

Table S10: DT outlet against the total liquid-water mobility factor F at $K_\ell = 0.32$ mol/(m^2 s); every other input as in Table S11. The last three rows give the maximum normalized water (W), hexane (H) and energy (U) defects of the independent reconstruction from dry-shell formation to exit (acceptance 10^{-10}).

Mobility factor F	100	1000 (earlier)	10 000
Moisture, % wet	17.62	18.20	18.27
Hexane, mg/kg dry	158.0	135.2	132.5
Hexane, ppm wet	130.1	110.6	108.3
Mean T, °C	105.01	104.98	104.97
Film consumed, s	702.5	624.5	614.0
W defect	7.1×10^{-12}	4.2×10^{-11}	2.6×10^{-11}
H defect	2.1×10^{-12}	8.6×10^{-13}	2.8×10^{-12}
U defect	2.7×10^{-12}	6.2×10^{-12}	4.4×10^{-12}

Construction of the contact conductance. The liquid contact conductance represents the resistance to liquid entry between the external water film and the connected particle matrix. It keeps the surface conductance finite when the shell forms, where the main-text law gives $G_\ell \to K_\ell$ as $d \to 0$. A conditional rate scale is constructed from hot-water uptake by soybean meal (Hemmingsen et al., 2008): the coarse sieve fraction (above 0.5 mm, mean 1.43 mm; 25 g fat per kg dry matter; initial water 11.8%, that is $W_0 = 0.134$ kg/kg dry) immersed at about 80 °C. Its rehydration index is near its later level at the first sample, 90 s, with some subsequent change. The construction assumes a sphere of radius $R = 0.715$ mm, entry through the surface alone, the surface at the film's state (W_cap) and the retention law of the main text. The water balance $(\rho_\mathrm{dm} V/M_\mathrm{w})\dot{W} = -AK_\ell[\psi(W) - \psi(W_\mathrm{cap})]$ with $\psi(W) = (1 - A_1/W)/A_2$ integrates in closed form to

$$t = \frac{\rho_\mathrm{dm} R A_2}{3 M_\mathrm{w} A_1 K_\ell}\, W_\mathrm{cap}\left[(W_0 - W_1) + W_\mathrm{cap} \ln \frac{W_\mathrm{cap} - W_0}{W_\mathrm{cap} - W_1}\right],$$

and requiring $W_1 = W_\mathrm{cap} - 0.01$ kg/kg at $t = 90$ s gives $K_\ell = 0.253$ mol/(m^2 s) at 80 °C. Scaling the entry rate with the inverse viscosity of water from 80 to 100 °C (a factor of 1.26) gives 0.318, adopted as 0.32. Changing the assumed end-point tolerance to 0.02 or 0.005 kg/kg below capacity gives 0.21 or 0.43. Reading the measurement at 20 °C with the same construction and viscosity

scaling (a factor of 3.55) gives 0.90, so the adopted value is the lower, less extrapolated, of the two readings.

The mapping assumes a spherical sieve diameter, the chosen retained-water endpoint and inverse-viscosity scaling. The source measures wet filter-cake mass per dry-solid mass after about 150 s of vacuum drainage. Its water includes particle, interstitial and surface contributions, which are not separated; the cake uptake also exceeds the model's retained capacity. Thus 90 s is not an isolated intraparticle equilibration time or a strict bound on it. The resulting K_ℓ is a conditional scale estimate, transferred to DT meal without fitting its outlet.

With the adopted liquid-water mobility, the resistance length $\mathcal{L}_\ell/K_\ell$ is 0.262 mm, 0.175 times the DT radius. At $d = R_\mathrm{p}$ the series conductance is $G_\ell = 0.0476$ mol/(m^2 s), and the contact term's share of the series resistance is 0.149. The contact-only time $\tau_K = \rho_\mathrm{dm} R_\mathrm{p} A_2 W^2/(3 M_\mathrm{w} K_\ell A_1)$ gives 56 s at $W = 0.20$, and the closed-form fill of a DT sphere limited by contact alone, from the feed loading 0.124 to $W_\mathrm{cap} - 0.01$, takes 153 s (491 s in the earlier reference case 0.10). These are scale estimates; radial gradients, pore-gas exchange, the changing surface state, the film's water stock and the capacity all remain active in the calculation.

Sensitivity to the contact conductance. Table S11 lists completed DT marches at four liquid contact conductances with every other input held constant, the earlier reference case 0.10 among them. The outlet moisture is bounded by the condensate supply and the retention capacity: the external water film is consumed before the exit in every case, and a fourfold change in K_ℓ moves the outlet moisture by 1.3 percentage points and the wet-basis hexane by 43 ppm. The conductance sets how long the film persists, not how much water the particle finally holds.

Table S11: DT outlet against the liquid contact conductance; total mobility factor 1000, two radial cells, regime-C maximum step 5 s, exit at 1766.5 s. Wet-basis values exclude the external film, which is zero at the exit in every case. Ordinary regime-B intervals were 0.1 s, except 0.2 s in the earlier 0.10 case.

K_ℓ, mol/(m^2 s)	0.05	0.10	0.20	0.32 (reference)
Moisture, % wet	16.68	17.40	17.94	18.20
Hexane, mg/kg dry	194.5	165.3	144.8	135.2
Hexane, ppm wet	162.0	136.5	118.8	110.6
Mean T, °C	105.09	105.03	104.99	104.98
Film consumed, s	816.8	733.6	662.4	624.5

Earlier constant-property events and refinement. The earlier DT calculation uses the feed history up to 492.257073 s, effective radius 1.5 mm, two radial cells, the prescribed bath and the existing retention capacity. The moving-front calculation begins with a 0.01-s shell-formation interval, advances regime B in 0.1-s intervals, shortened where events require, and resolves grid-face arrival and departure before core extinction at 503.560810 s. External water stays positive throughout regime B, so the zero-film regime-B branch, which is not implemented, is not exercised (Sec. S11). Regime C advances the stationary particle and the film together to the film's depletion at 624.548745 s and then to the exit. Table S12 gives the regime-C step refinement, performed in the earlier reference case $K_\ell = 0.10$ from the same moving-front end point; the 5-s maximum step is used in every reported march. The extra digits resolve the numerical sensitivity; the main text uses rounded values.

The numerical recoveries for these earlier contact and mobility cases, including the single accepted water-row exception at factor 10 000, are specified together in Sec. S2.2.

Table S12: Regime-C timestep sensitivity of the DT particle with liquid-water imbibition, $K_\ell = 0.10$ mol/(m^2 s).

Quantity	Maximum step 10 s	Maximum step 5 s
Moisture, % wet basis	17.3813	17.4033
Hexane, ppm wet basis	137.222	136.474
Mean temperature, °C	105.0329	105.0314
External-water depletion time, s	735.395	733.575

Earlier constant-property independent balances. An independent reconstruction of the earlier reference march, from dry-shell formation to exit, evaluates water (W), hexane (H) and energy (U) in the particle and external water film at every accepted step. The maximum normalized W/H/U defects are 4.24×10^{-11}, 8.61×10^{-13} and 6.16×10^{-12}, below the 10^{-10} acceptance limit; the regime-B segment alone gives 1.05×10^{-11}, 8.61×10^{-13} and 3.94×10^{-12}. The same reconstruction stays below 4.2×10^{-11} for the other three conductances of Table S11. The unchanged feed history keeps its earlier independent check, which stays below 3.3×10^{-13} up to dry-shell formation. Component partial enthalpies count each energy transfer once. These checks quantify component and energy conservation in the reported particle calculation. Complete-history mesh and time-step refinement (Sec. S11) and independent identification of the material coefficients provide complementary evidence about accuracy and physical prediction.

S13 Temperature-dependent properties and prescribed sensitivity

The nominal pore and oil laws, source definitions and applicability assumptions are stated with main Eqs. 12 and 31–32. The pore coefficient is evaluated at each common face temperature and pressure. Retained-water mobility, retention, conductivity, geometry and baths remain fixed; no moisture target or outlet is refitted.

Reference-magnitude sensitivity. The reference magnitudes 2, 4 and 6×10^{-10} m^2/s are prescribed alternatives, not a measured uncertainty interval. Each case starts from the original feed; the DT regime-A feed is common because binary pore transport is inactive there. Accepted-state restarts retain their own property law and cumulative ledgers. The DT histories pass independent water, hexane and energy reconstruction from shell formation to exit at 10^{-10}; complete Faner histories pass the independently accumulated exterior-rate audit at 4×10^{-10}.

Table S13: Complete DT histories under the temperature-dependent laws. Reference diffusivity in 10^{-10} m^2/s at 105°C and one atmosphere; liquid contact conductance 0.32 mol/(m^2 s), total mobility factor 100.

$D_{b,\mathrm{ref}}$	Moisture % wet	Hexane ppm wet	Hexane mg/kg dry	T °C	Film ends s
2	17.3193	145.480	175.985	104.9340	703.666
4	17.4618	138.389	167.695	104.9648	703.260
6	17.6151	130.274	158.154	105.0085	702.554

The endpoint span is 0.296 moisture percentage points; increasing the reference from 2 to 6 reduces residual hexane by 10.45%. This quantifies sensitivity within the prescribed interval; it does not establish the true material coefficient.

The Faner falling interval and outlet change very little across these cases. The adaptive histories use different accepted intervals and documented recovery controls. Without refinement of

Table S14: Complete Faner histories with the same prescribed reference diffusivities, four radial cells and the fixed constant-rate exposure multiplier.

$D_{b,\mathrm{ref}}$	Core ends s	$0.20 \rightarrow 0.10$ s	$X_{\mathrm{h}}(240)$ kg/kg dry	$T(240)$ °C
2	87.647	17.745	0.001606	114.351
4	87.362	17.746	0.001606	114.354
6	87.824	17.745	0.001606	114.349

the complete histories, the subsecond differences in core-ending time cannot be separated from numerical-history effects.

Temperature-shape comparison. At the same reference magnitude 4 and with the oil law unchanged, a complete constant-D_b DT history gives 138.286 ppm wet hexane and 17.4648% moisture. Applying the Fuller shape changes them by +0.075% and -0.00291 percentage points. The reference magnitude therefore matters more here than this prescribed temperature shape.

The temperature-shape hexane difference is smaller than the observed regime-C timestep change below; it should not be interpreted as a resolved physical improvement.

Regime-C timestep check. With the same accepted regime-B checkpoint, the nominal laws, factor 100 and $K_\ell = 0.32$ mol/(m^2 s), reducing the maximum regime-C step from 5 to 2.5 s changes wet-basis moisture by +0.01170 percentage points, residual hexane by -0.295%, temperature by -0.00063 K and first recorded film depletion by -1.079 s. Both histories pass the independent component and energy audit at 10^{-10}. This checks regime-C timestep sensitivity; full-history mesh and timestep convergence remains open.

Oil heat capacity and caloric consistency. The oil fit and its analytic energy integral are given in main Eqs. 31–32. At 105°C it gives 2158.45 rather than 2000 J/(kg K). The reference anchoring cancels the oil correction for Faner; at the DT oil fraction 0.013740458, dry-matrix capacity at 105°C changes from 2318.8621 to 2317.9313 J/(kg K), or −0.04014%. This is a capacity comparison, not a bound on the nonlinear trajectory error. Regenerated DT feed reaches shell activation 2.46 microseconds earlier than the former constant-oil feed. Its independent exterior-rate audit is stated in Sec. S11. The physical property adapter supplies the same capacity to wet and dry temperature derivatives; stored constant capacity fields remain numerical reference scales.

Numerical recoveries and accepted-state controls for these cases are collected in Sec. S2.2. Earlier contact, mobility-factor and mesh comparisons retain their constant-property scope. The new laws remain conditional material assumptions, not validated binary transport or an equipment model.

S14 Governing and supporting equations with their source basis

The following tables expand the main-text equation map. They are a reading guide to the equations already defined in the two documents, with their literature basis or derivation stated alongside. A citation to a general framework does not identify the present coupled model or its coefficients as a published or experimentally validated law. “Present closure” denotes the formulation adopted here; a literature-informed coefficient can remain an engineering assumption.

Throughout the tables, $\mathcal{D}_r(F) = r^{-2}\partial_r(r^2F)$, $s = s$, $z = (s/R_{\mathrm{p}})^3$, and $[\cdot]^{\mathrm{d}}_{\mathrm{w}}$ is the dry-side value minus the wet-side value. The region label w denotes the liquid-hexane core; component w denotes water. C_k includes all local forms of component k per particle volume, u is total stored energy

per volume, and $J_E = q + \sum_k \bar{h}_k j_k$. Local j_k is molar; surface-film j_k is mass-based and converts through M_k. y_k and Y_k are gas mole and mass fractions, respectively. Fluxes are outward-positive. Repeated state arguments are omitted; the referenced equation or section supplies the full definitions and domains.

Table S15: Solved balances, regional inventories and interface conditions.

Equation / block	Compact expression	Physical role / scope	Source / derivation
Local components (2)	$\partial_t C_k^\alpha + \mathcal{D}_r(j_k^\alpha) = 0$	Conserve water and hexane in either smooth region; phase redistribution has no separate source.	Component conservation, with the present regional storage and fluxes.
Local energy (3)	$\partial_t u^\alpha + \mathcal{D}_r(J_E^\alpha) = 0$ $J_E = q + \sum_k h_k j_k$	Evolve temperature while counting each stream's enthalpy once; dry meal and oil are immobile.	Energy conservation on one component datum; present coupling.
Dry-shell storage (4)	$C_k^{\mathrm{d}} = \varepsilon_g c y_k + \rho_{\mathrm{dm,p}} W_k / M_k$	Add pore-gas and retained inventories without including gas in W_k.	Present phase-volume accounting, closed by the property laws in Table S16.
Core material label (5)	$\mathrm{D}_s(\rho_{\mathrm{dm,p}} X_{\mathrm{h,w}}^a)/\mathrm{D}t = 0$	Preserve activation loading in surviving wet material; no relative core hexane flux.	Present historical-core assumption; $X_{\mathrm{h},w}^a = X_c(\theta_a)$ at activation.
Wet-core energy (6)	$e_{\mathrm{w}} = h_{\mathrm{dm}} + X_{\mathrm{w}} h_{\mathrm{w,l}} + X_{\mathrm{h,w}}^a u_{\mathrm{h,l}} - B_{\mathrm{h}}(1, T, w_o)$	Store matrix, water, historical hexane and binding energy per dry-meal mass.	Present composite caloric closure; pure-fluid and binding sources in Table S16.
Moving volumes (13)–(14)	$\frac{\mathrm{d}}{\mathrm{d}t}\int_V b\,\mathrm{d}V = -\int_{\partial V}(F_b - \dot{r}_b b) n_r\,\mathrm{d}A$ $b = C_k, u$; $F_b = j_k, J_E$	Include material swept by the moving radial boundaries. Here $\dot{r}_b$ is their radial speed and $n_r = \pm 1$.	Moving-control-volume conservation, specialized to a sphere in the main text.
Front jumps (15)–(16)	$[j_k - \dot{s} C_k]_{\mathrm{w}}^{\mathrm{d}} = 0$ $[J_E - \dot{s} u]_{\mathrm{w}}^{\mathrm{d}} = 0$	Match water, hexane and energy across the sharp front, with stored donor states in swept terms.	Rankine–Hugoniot conservation (Crank, 1984); present phase closures.
Front speed (17)	$\dot{s} = -\dfrac{[q] + \bar{h}_{\mathrm{w,d}} j_{\mathrm{w,d}} - \bar{h}_{\mathrm{w,w}} j_{\mathrm{w,w}}}{\mathcal{E}_\Gamma}$ $\mathcal{E}_\Gamma = \bar{h}_{\mathrm{h,d}}(C_{\mathrm{h,d}}^\Gamma - C_{\mathrm{h,w}}^{\mathrm{b}}) - (u_{\mathrm{d}}^\Gamma - u_{\mathrm{w}}^{\mathrm{b}})$	Heat-fed recession derived from both jumps and zero relative wet-side hexane flux; no total-vapour pressure resistance.	Present elimination of the hexane flux. Eq. (S1) explains $\mathcal{E}_\Gamma$.

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Table S15 (continued)

Equation / block	Compact expression	Physical role / scope	Source / derivation
Sequential drainage (18)	$X_f \geq 0,\ R_p - s \geq 0$ $X_f(R_p - s) = 0,\ \dot{s} \leq 0$	Exhaust external free hexane before recession; admit primary drainage only.	Present regime and admissibility conditions; violations are rejected.
Shared swept volume (33)	$Q_\Gamma = V_{R_p}(z_{\mathrm{new}} - z_{\mathrm{old}})/\Delta t$	Use one signed volume-change rate in component, energy and front equations.	Present finite-volume geometric conservation.
Retained-water capacity (34)	$0 \leq W_{\mathrm{cap}} - W \ \perp\ \lambda \geq 0$	Enforce capacity as an active set; $\lambda = 0$ below capacity.	Present constraint using the source-domain retention capacity; Sec. S1.2.

Table S16: Transport, retention, surface exchange and thermophysical closures.

Equation / block	Compact expression	Physical role / scope	Source / derivation
Binary pore exchange (7)	$J_{\mathrm{w}} = -J_{\mathrm{h}} = -L^{\mathrm{eff}}_{\mathrm{wh}} \nabla_{\mathrm{comp}} \Xi$ $\Xi = (\mu_{\mathrm{w}} - \mu_{\mathrm{h}})/(RT)$ $L^{\mathrm{eff}}_{\mathrm{wh}} = c y_{\mathrm{w}} y_{\mathrm{h}} D_b$	Separate the two vapours relative to molar-average motion; the mobility vanishes at purity.	Maxwell–Stefan framework (Krishna & Wesselingh, 1997); present nonideal force evaluation.
Retained-water and total gas flux (8)	$N_{\mathrm{w,ret}} = -L^{\mathrm{eff}}_{\mathrm{w,ret}} \nabla_{\mathrm{comp}}(\mu_{\mathrm{w,ret}}/RT)$ $N_i^{\mathrm{g}} = y_i N_t + J_i$	Supply a distinct water path; recover N_t algebraically from continuity. Independent Soret terms are omitted.	Present two-path closure; retained mobility construction in Sec. S12.
Nonisothermal face force (9)	$G_{\mathrm{comp}}(\Xi) = \dfrac{1}{d}\displaystyle\int_0^1 \left.\dfrac{\partial \Xi}{\partial y_{\mathrm{h}}}\right\vert_{T(\lambda),P} \dfrac{\mathrm{d}y_{\mathrm{h}}}{\mathrm{d}\lambda}\,\mathrm{d}\lambda$	Integrate only the composition force along the actual interface reconstruction; ordinary faces use a common T.	Present reconstruction under the negligible-Soret approximation.
Pore diffusivity (12)	$D_b = 4.0 \times 10^{-10} \left(\dfrac{T}{378.15\,\mathrm{K}}\right)^{1.75} \dfrac{101325\,\mathrm{Pa}}{P}$ $[D_b] = \mathrm{m}^2/\mathrm{s}$	Nominal pore-effective law; assumes constant hindrance. Prescribed magnitudes (2–6) $\times 10^{-10}$ m^2/s are a sensitivity range.	Shape: Fuller, Schettler, and Giddings (1966). Scalar scale: Cardarelli et al. (2002). Binary magnitude unvalidated; Sec. S13.

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Table S16 (continued)

Equation / block	Compact expression	Physical role / scope	Source / derivation
External water and hexane (20)–(22)	$\dot{L}_{\mathrm{w}} = j_{\mathrm{w},p} - j_{\mathrm{w},g}$ $j_{\mathrm{h},p} = j_{\mathrm{h},g}$ $\mathrm{d}(L_{\mathrm{w}} u_{\mathrm{w},l})/\mathrm{d}t = J_{E,\mathrm{p}} - J_{E,\mathrm{g}}$	Store condensate separately. Water appearance/depletion conserves stock and energy; hexane passes without added resistance.	Present surface balances and declared hexane-transparent water-film limit.
Gas-film components (23)	$j_{\mathrm{h},g} = Y_{\mathrm{h},s} j_t + K_f(Y_{\mathrm{h},s} - Y_{\mathrm{h},b})\Phi(j_t/K_f)$ $j_{\mathrm{w},g} = j_t - j_{\mathrm{h},g}$	Finite-flow ideal binary mass-coordinate film, using equation-of-state phase traces.	Analytic film closure adopted here; heat/mass exposure convention in Sec. S7.2.
Ackermann correction (24)	$\Phi(x) = x/(e^x - 1)$, $\Phi(0) = 1$ $\beta = \dfrac{j_{\mathrm{w},g} c^f_{p,\mathrm{w}} + j_{\mathrm{h},g} c^f_{p,\mathrm{h}}}{h_Q}$	Both signed component capacity fluxes enter, including opposing flows; declared band $\lvert\beta\rvert \le 3$.	Adopted analytic finite-flow correction; the wider film-law band remains an assumption.
Gas-film energy (25)	$q_{\mathrm{in}} = h_Q \Phi(\beta)(T_{\mathrm{bulk}} - T_s)$ $J_{E,\mathrm{g}} = -q_{\mathrm{in}} + \sum_k (j_{k,\mathrm{g}}/M_k)\bar{h}_k(T_s)$	Combine sensible heat and stream enthalpy on the same datum; no added latent source.	Present energy coupling; bed coefficient from Faner (2008).
Finite liquid entry (26)	$G_\ell = (d/\mathcal{L}_\ell + 1/K_\ell)^{-1}$ $N_\ell = -G_\ell(\psi_s - \psi_p)$, $\psi = \ln a_{\mathrm{w,ret}}$	Contact and interior resistances in series; active only in a liquid-connected dry shell. Donor energy follows the water.	Present effective closure. Mobility and contact constructions in Sec. S12 and Table S17.
Finite film supply (27)	$j_{\ell,\mathrm{in}} \le \max\big(0, L_{\mathrm{w},old}/\Delta t + j^{\mathrm{base}}_{\mathrm{w},p} - j_{\mathrm{w},g}\big)$	Bound inward addition by stored and newly supplied water; debit film and credit particle equally.	Present discrete stock-availability condition.
Native-meal hexane retention (28)	$W_{\mathrm{native}} = \dfrac{X_m C x}{(1-x)(1-x+Cx)}$ $x = K(T) a_{\mathrm{h}}$	GAB total-meal isotherm; the reference sample already includes residual oil.	Soybean-meal fit of Cardarelli and Crapiste (1996); temperature-dependent C and K.
Oil anchoring (29)	$W_{\mathrm{h}} = \alpha_o W_{\mathrm{h,native}} + \beta_o q_o$ $\alpha_o = (1 - w_o)/(1 - w_{o,\mathrm{ref}})$ $\beta_o = (w_o - w_{o,\mathrm{ref}})/(1 - w_{o,\mathrm{ref}})$ $q_o = 0.9635 a_{\mathrm{h}}^{2.7036}$	Adjust off-reference oil content without duplicating reference oil. Use the same weights for sensible heat.	Present reference-anchored composite construction; source and comparison convention in Sec. S1.2.

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Table S16 (continued)

Equation / block	Compact expression	Physical role / scope	Source / derivation
Retained water (30)	$W_{\mathrm{w}} = A_1/[1 + A_2 \ln(1/a_{\mathrm{w}})]$ $\ln a_{\mathrm{w}} = (1 - A_1/W_{\mathrm{w}})/A_2$	Temperature-independent nominal isotherm. Positive low-water extension is declared; no lower water floor.	Modified Luikov fit (Luikov, 1975; Luz, Sousa, Jorge, & Paraíso, 2006; Pixton & Warburton, 1975); source limits in main Sec. 2.8.
Oil heat capacity (31)	$c_{p,\mathrm{o}}(T) = 1024 + 3T$ J/(kg K)	Soybean-oil sensible capacity; T in kelvin, reported range 299.15–433.15 K.	Rounded linear fit, Table 7 of Garcia Rojas, Coimbra, and Telis-Romero (2013).
Dry-matrix sensible energy (32)	$\Delta u_{\mathrm{dm}} = \alpha_o c_{p,\mathrm{native}}(T - T_r) + \beta_o[1024(T - T_r) + 1.5(T^2 - T_r^2)]$	Analytic integral of the same capacity used for the temperature derivative. T_r denotes the solid reference T_s of the full equation.	Integration of the oil fit with present reference anchoring; oil correction cancels at $w_o = 0.0195$.
Fluid properties and equilibrium Sec. 2.8	$f_{i,\mathrm{g}} = f_{i,\mathrm{l}}$ at an admitted interface $\bar{h}_{i,\mathrm{g}} = h_{i,\mathrm{g}}^{\mathrm{ideal}} + \bar{h}_{i,\mathrm{g}}^{\mathrm{res}}$	Close liquid density, fugacities and caloric values. Binary gas uses a second-virial mixture with a cross term.	Hexane: Span and Wagner (2003a, 2003b); water: IAPWS (2018); Wagner and Pruß (2002); mixture: Plyasunov and Shock (2003).
Hexane binding energy Sec. S1.1	$\psi_{\mathrm{h}} = \dfrac{RT}{M_{\mathrm{h}}} \displaystyle\int_0^W \ln a_{\mathrm{h}}(\xi, T)\, \mathrm{d}\xi$ $B_{\mathrm{h}} = T\partial_T \psi_{\mathrm{h}} - \psi_{\mathrm{h}}$	Derive retained-hexane calorics from the same retention law; no independently fitted sorption-heat source.	Present thermodynamic construction from the GAB activity law (Cardarelli & Crapiste, 1996).

Table S17: Supporting correlations, coefficient constructions and diagnostics. The final three rows belong to the separate historical Fickian comparison.

Equation / block	Compact expression	Physical role / scope	Source / derivation
Ideal activity reference (1)	$a^* = \min(1, P/P_{\mathrm{sat}}^{\mathrm{h}}(T))$	Diagnostic only. Actual binary phase equilibrium uses equation-of-state fugacities.	Ideal pressure-feasibility limit; present regime-ordering comparison.

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Table S17 (continued)

Equation / block	Compact expression	Physical role / scope	Source / derivation
Pressure-drop scale (10)	$\Delta P \simeq (N_t RT/P)(\mu_g/B)(R_p - s)$	Assess the isobaric assumption. This estimate adds no Darcy resistance to the solved total-flux law.	Darcy scaling with declared illustrative permeability and vapour-flux bounds.
Ideal binary limit (11)	$J_{\mathrm{h}} = -cD_b\partial_r y_{\mathrm{h}}$ $N_{\mathrm{h}}^{\mathrm{g}} = y_{\mathrm{h}} N_t + J_{\mathrm{h}}$	Interpret exchange versus carried flux. Computations retain the nonideal chemical force.	Ideal fixed-(T, P) limit of the binary law (Krishna & Wesselingh, 1997).
Caloric consistency (19)	$\partial R_E/\partial c_{\mathrm{w}} = R_{\mathrm{w}}$	Datum-shift identity for the discrete interface residuals; a consistency test, not an evolution equation.	Present consequence of consistent component and energy accounting.
Geometric comparator (35)	$s = R_p[(X_{\mathrm{h}} - X_e)/(X_c - X_e)]^{1/3}$	Loading-to-radius comparison under uniform core/shell assumptions. The solved front instead follows the jumps.	Geometric reconstruction using source-based loadings; Sec. 4.1.
Water-only temperature (36)	$a_{\mathrm{w}}(W_{\mathrm{w}})P_{\mathrm{sat}}^{\mathrm{w}}(T^*) = P$	Conditional equilibrium diagnostic. The coupled binary energy balance determines the actual temperature.	Water activity plus vapour-pressure relation; present interpretation.
Bed heat-transfer correlation Sec. S1.2	$\mathrm{Nu}_\varepsilon = 0.6941\mathrm{Re}_\varepsilon^{0.579}\mathrm{Pr}^{1/3}$ $\mathrm{Re}_\varepsilon = \mathrm{Re}/(1-\varepsilon)$ $\mathrm{Nu}_\varepsilon = \mathrm{Nu}\varepsilon/(1-\varepsilon)$	Holder voidage ε differs from particle porosity. Actual exposure and heat/mass scaling are separate assumptions.	Thesis convention (Faner, 2008); its 0.6941 prefactor is used.
Size-distribution weights Sec. S5	$q_c = p_c d_c^3 / \sum_b p_b d_b^3$ $\overline{X}_k = \sum_c q_c X_{k,c}$	Convert number probabilities to dry-mass weights for equal-density spheres; requires complete class histories.	Geometric mass weighting; assumed distribution consistent with Faner et al. (2019).
Constant-rate slope Sec. S7.2	$r_c = -\dfrac{\sum_j (t_j - \bar{t})(X_j - \bar{X})}{\sum_j (t_j - \bar{t})^2}$	Least-squares slope of four digitized constant-rate observations; these data remain listed in the referenced section.	Observations: Faner et al. (2019), Fig. 2. Present boundary-exposure construction.
Exposure-factor construction Sec. S7.2	$r_i(f) = \dfrac{3fh_{Q,i}^0}{\rho_{\mathrm{dm}} R_i c_{p,v}} \ln\left(1 + \dfrac{c_{p,v}(T_g - T_{\mathrm{boil}})}{\Delta h_{\mathrm{vap}}}\right)$	Match the constant-rate duty, then carry the exposure factor into the particle boundary. Falling-rate data are reserved for comparison.	Present inversion of the Ackermann heat balance; source data from Faner et al. (2019).

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Table S17 (continued)

Equation / block	Compact expression	Physical role / scope	Source / derivation
Saved flux decomposition Sec. S10.1	$N_{\mathrm{h}}^{\mathrm{g}} = y_{\mathrm{h},up} N_t + J_{\mathrm{h}}$ $f_{\mathrm{motion}} = y_{\mathrm{h},up} N_t / N_{\mathrm{h}}^{\mathrm{g}}$ $f_{\mathrm{exchange}} = J_{\mathrm{h}} / N_{\mathrm{h}}^{\mathrm{g}}$	Signed diagnostic contributions; opposing terms can exceed 100%. These are not causal rate-control fractions.	Present identity applied to saved native surface rates.
Reference water mobility Sec. S12	$D_{\mathrm{app}} = D_{\mathrm{app,pore}} + g_{\mathrm{ret}} \mathcal{L}_{\mathrm{w,ret}}$	Original local linearization at a declared reference state; retain the resulting mobility when changing D_b.	Present construction using apparent response from Touffet et al. (2026); material/state transfer assumed.
Retained-path diffusivity Sec. S12	$D_{\mathrm{ret}} = M_{\mathrm{w}} \mathcal{L} A_1 / (\rho_{\mathrm{dm}} A_2 W^2)$	Isothermal uncapped Luikov limit. Liquid-connected shell uses total mobility factor 100; the added part is 99.	Derived from the activity derivative; factor scale informed by Künzel (1995), transferred from other porous solids.
Contact-only fill time Sec. S12	$t = \frac{\rho_{\mathrm{dm}} R A_2 W_{\mathrm{cap}}}{3 M_{\mathrm{w}} A_1 K_\ell} \left[W_0 - W_1 + W_{\mathrm{cap}} \ln \frac{W_{\mathrm{cap}} - W_0}{W_{\mathrm{cap}} - W_1} \right]$	Uniform-sphere scale used to construct K_ℓ, with assumed endpoint and inverse-viscosity transfer. Actual march retains gradients and finite stock.	Present integration using the first 90-s uptake sample of Hemmingsen et al. (2008); not an identified DT contact coefficient.
Historical Fickian shell (S2)	$\partial_t [\varepsilon_g c_{\mathrm{h,g}} + \rho_{\mathrm{dm,p}} W_{\mathrm{eq}}] = \mathcal{D}_r (D_{\mathrm{eff}} \partial_r c_{\mathrm{h,g}})$	Separate constitutive description for the 2008 falling-rate comparisons; not the nominal binary pore model.	Storage-corrected scalar diffusion comparison; Cardarelli (1998); Faner (2008).
Historical diffusivity fit (S3)	$\ln D_{\mathrm{eff}} = \ln A - E_a/(R_g T) + \beta \ln(X_{\mathrm{h}} / X_{\mathrm{ref}})$	Arrhenius/loading fit for the separate Fickian comparator. This is not the nominal $D_b(T, P)$ law.	Regression on independent uptake coefficients (Cardarelli, 1998); Sec. S4.
Historical supply diagnostic (S4)	$\Lambda = \frac{k_{\mathrm{dry}} (T_g - T_{\mathrm{int}}) / \Delta h_{\mathrm{vap}}}{D_{\mathrm{eff}} [c_{\mathrm{sat}}(T_g) - c_s]}$	Compare heat and diffusive supply at a fixed state in the separate Fickian description; no added resistance in the DT march.	Present scale comparison, defined in Sec. S6.

S15 Reproduction inventory and accessibility

The release matching this submission is at `https://github.com/tomas3svoboda/dt_particle_model`, submission snapshot `submission-2026-10-15`, with a SHA-256 manifest of every file. It holds the particle-model source code; the scripts that produced the two particle histories and the sensitivity calculations of Sec. S12; their run records, including the saved states from which each segment starts and the numerical-recovery records; the prescribed DT bath as a declared input file; the digitized readings of the 2019 article; the scripts that rebuild the three computed figures (two in the main text, one in this supplement) and check the printed numbers against their records; and the LaTeX sources and PDFs of both documents. Files keep their relative paths, and the release's README gives the commands. The equipment model that computed the prescribed DT bath is not part of the release, so the bath cannot be regenerated from it. Some early steps of the Faner calculation and the $K_\ell = 0.10$ case were run with earlier revisions of the calculation scripts; each step record stores the hashes of the scripts it used. The release holds the records behind the figures and the headline numbers of the main text, and the earlier records that the number check reads; the number check binds the printed numbers of both documents to the released records and names those that rest on records the release does not hold. Such records, among them the authors' working notes, the derivation records and the independent balance record of the DT feed history, are available from the corresponding author on request. Other digitized readings are not in the release; they are shared where the publishers' terms permit. The authors' code is under MIT terms, which do not cover the third-party publications or their digitizations. A preservation archive with a persistent identifier is being prepared.

Nomenclature of the supplement

Symbols below appear in this supplement and not in the main-text nomenclature. Every other symbol keeps the meaning given there. The Ackermann blowing argument is β_{A} in both documents; β here is only the exponent in the diffusivity fit.

a_{h}^{*} pressure-feasible n-hexane activity at a stated gas mole fraction, $\min(1, y_{\mathrm{h}}P/P_{\mathrm{sat}}^{\mathrm{h}})$; dimensionless

A pre-factor of the uptake diffusivity fit (S3); m^2/s

i, n component and time indices where used; dimensionless

C_0, K_0 reference constants of the native-meal GAB isotherm; dimensionless

d_{eq} equivalent particle diameter printed by Faner (2008); m

ΔH_C, ΔH_K GAB energy parameters, reported as $\Delta H_C/R$ and $\Delta H_K/R$; J/mol

E_{a} activation energy in the uptake diffusivity fit; J/mol

h_l, h_v specific enthalpy of liquid n-hexane and of n-hexane vapour; J/kg

r_1, r_2 inner and outer radii of the spherical control volume; m

r_{c} measured constant drying rate, loading lost per unit time; 1/s

R_{d} radius of the declared volume-to-surface sphere; m

T_{b} boiling temperature of n-hexane at the bed pressure; K

u_l, u_v specific internal energy of liquid n-hexane and of n-hexane vapour; J/kg

v_l specific volume of liquid n-hexane; m^3/kg

W_{ref} lower anchor of the retained-water evidence; kg/kg

$X_{\mathrm{e,sh}}$ coupled-pore shell sorbate in the lumped desolventizer front relation, at the shell mean temperature and admitted near-interface gas composition; kg/kg

X_{ref} reference loading in the uptake diffusivity fit; kg/kg

Δz sublayer thickness; m

β exponent of the loading term in the uptake diffusivity fit; dimensionless

ϵ emissivity in the radiative bed term; dimensionless

ε holder voidage in the source bed-film definition; not the particle porosity ε_{g}; dimensionless

Nu, Re, Pr Nusselt, Reynolds and Prandtl numbers of that definition; $\mathrm{Nu}_{\varepsilon}$ and $\mathrm{Re}_{\varepsilon}$ are the source's voidage-based forms; dimensionless

σ Stefan–Boltzmann constant; $\mathrm{W}/(\mathrm{m}^2\,\mathrm{K}^4)$

ψ sphericity; dimensionless

ψ_{h} sorption potential of retained n-hexane; J/kg

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